

Early to middle Eocene (50–43 Ma) Xianqian Basin in central Tibet: Archives of arid paleoenvironments and a southward plateau uplift trend

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ABSTRACT

Sedimentary basins provide valuable records of local surface uplift and paleoenvironment. While previous studies have inferred that the Qiangtang terrane in the Tibetan interior has been at high elevation since at least the Eocene, direct evidence from well-dated basins within the Qiangtang terrane remains scarce. Here, we present integrated stratigraphic, sedimentologic, paleontologic, and provenance data from the Maerguo Formation in the Xianqian Basin, southwestern Qiangtang. U–Pb dating of tuff zircons and lacustrine carbonate indicates that the Maerguo Formation was deposited between ca. 50 Ma and 43 Ma, establishing an Eocene record for central Tibet's paleoenvironmental, tectonic, and topographic evolution. The Maerguo Formation consists of >1900 m of well-exposed strata that comprise five distinct units from bottom to top: Em₁ (lower conglomerates); Em₂ (fluvial red beds); Em₃ (lower fine-grained lacustrine red beds); Em₄ (upper lacustrine unit); and Em₅ (upper conglomerates). Growth strata developed synchronously with a north-dipping thrust along the northern margin of the Xianqian Basin, while the southern part of the basin dips gently and lies unconformably above Mesozoic

strata. Multiproxy provenance data indicate sediment sources from surrounding sedimentary and volcanic rocks, with sediments derived from more localized metamorphic rocks and Jurassic granitoids to the north becoming dominant during the deposition of units Em₄ and Em₅ in the northern basin. The presence of abundant gypsum in Em₃, lacustrine carbonate in Em₃ and Em₄, and small, narrow, elongate leaf fossils in Em₄ indicates an arid environment. These findings indicate that the Xianqian Basin was a short-lived, arid, internally drained system that terminated well before the peak expansion of Cenozoic basins along the Bangong–Nujiang suture to the south. The integrated paleogeographic reconstruction suggests that surface uplift of the Tibetan Plateau progressed southward from its interior since the Eocene. This Tibetan uplift pattern may have been driven by lithospheric removal and associated crustal shortening and thickening, beginning in the central Qiangtang terrane.

1. INTRODUCTION

The Tibetan Plateau, the largest and highest plateau on Earth, has experienced a complex uplift history since the Mesozoic due to the successive suturing of multiple terranes (Dewey et al., 1988; Yin and Harrison, 2000; Wang et al., 2008a; Kapp and DeCelles, 2019; Spicer et al., 2021; Ding et al., 2022). Evidence indicates the presence of an Andean-type Gangdese magmatic arc >3 km in paleoelevation just before

the India–Asia collision in the southern Lhasa terrane of southern Tibet (Ibarra et al., 2023), and this high topography persisted into the early Cenozoic (Ding et al., 2014; Ingalls et al., 2018).

Farther to the north, the Bangong–Nujiang suture zone in central Tibet divides the Lhasa and Qiangtang terranes and hosts several Cenozoic nonmarine basins that record signals of early Cenozoic low elevation (Xu et al., 1985; Rowley and Currie, 2006; DeCelles et al., 2007; Wang et al., 2011; Wei et al., 2016; Han et al., 2019). Among these basins, the Lunpola–Bangoin–Nima basin in the east is well constrained to have been wet and at a low elevation during the Paleocene to Eocene (Wu et al., 2017; Su et al., 2019, 2020; Fang et al., 2020; Xiong et al., 2022; Li et al., 2022b, 2024b). The Gerze basin to the west was suggested to be near sea-level elevation during the Eocene (Wei et al., 2016), though supporting evidence has been argued to be insufficient (Li et al., 2024a).

By contrast, paleoelevation estimates are less well constrained for the Qiangtang terrane. Though low $\delta^{18}\text{O}$ values from strata of poorly constrained depositional age (e.g., some time between ca. 51 Ma and 28 Ma) suggest >5000 m (Xu et al., 2013) or >2830 m of paleoelevation (Shen et al., 2023), oxygen isotope records in Tibet are prone to diagenetic alteration and thus require careful assessment (Quade et al., 2020; Li et al., 2024a, 2024c). Other evidence indirectly suggests that the Qiangtang terrane may have been uplifted as an orogenic plateau during the Eocene, including (1) weakly deformed flat-lying Eocene to Oligocene volcanic rocks (Law

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and Allen, 2020) and (2) regional exhumation of rocks to <3 km crustal depth by ca. 45 Ma (Rohrmann et al., 2012). However, these lines of evidence alone do not necessitate the existence of a high-elevation plateau.

Cenozoic sedimentary basins in the Qiangtang terrane provide an opportunity to better reconstruct its paleoenvironment and paleotopographic history. Previous studies have grouped Cenozoic strata across the Qiangtang terrane into the lower coarse-grained Kangtuo

Formation and the upper fine-grained Suonahu Formation, interpreted to have been deposited in alluvial fans and lacustrine environments, respectively (Xia and Liu, 1997). However, only a few Paleogene radioisotopic ages are available for these strata (Xu et al., 2013; Wang et al., 2019; Zhang et al., 2024), and systematic studies of these units at the scale of individual basins are still lacking.

In this paper, we report the results of an investigation of the Maerguo Formation from

the Eocene Xianqian Basin in the southwestern Qiangtang terrane (Fig. 1). Our detailed measured stratigraphic sections of the Maerguo Formation show that it consists of an ~2-km-thick package of coarse- to fine-grained clastic, carbonate, gypsum, and volcanoclastic strata. A major portion (472 m) of the upper Maerguo Formation consists of deep-water lacustrine gravity-flow deposits. Geologic mapping, multiproxy sediment provenance data, and fossils further illuminate the paleoenvironmen-

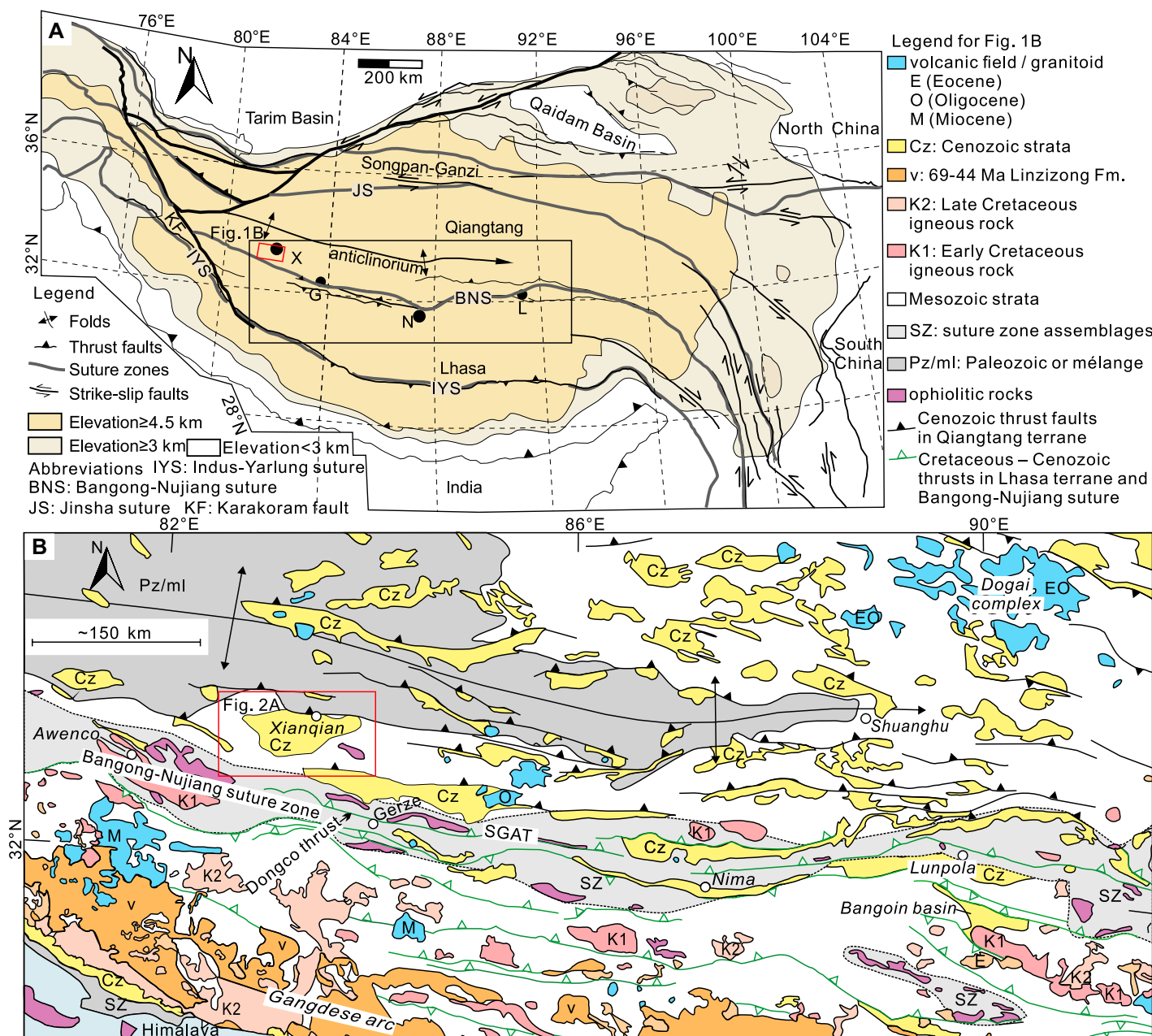


Figure 1. Regional geological map showing the location of the Xianqian Basin. (A) Tectonic framework of the Tibetan Plateau (after DeCelles et al., 2011). The Xianqian Basin (X) is located in central western Tibet. G—Gertze; N—Nima; L—Lunpola. (B) Geologic map of central-southern Tibet after Kapp and DeCelles (2019). SGAT—Shiquanhe-Gerze-Amdo thrust system.

tal, paleogeographic, and tectonic evolution of the Xianqian Basin. The data we present here indicate that the 50–43 Ma Maerguo Formation was deposited in the footwall of a coeval north-dipping thrust fault and was located in a region much drier and likely higher in elevation than previously documented contemporaneous basins within the Bangong-Nujiang suture zone to the south (Gerze, Nima, and Lunpola Basins). The stratigraphic architecture, syncontractional setting, abundance of tuffs and tuffaceous breccias in the upper part of the Maerguo Formation, and the relatively short-lived nature of the Xianqian Basin are broadly consistent with a basin-forming mechanism influenced by integrated upper-crustal shortening and magmatism related to removal of lower lithosphere.

2. GEOLOGIC BACKGROUND

The Qiangtang terrane, bounded by the Jinsha suture to the north and the Bangong-Nujiang suture to the south, constitutes the middle of the physiographic Tibetan Plateau (Fig. 1A). The terrane-scale eastward-plunging Qiangtang anticlinorium exposes early Paleozoic pan-African granitoid basement, Triassic high-pressure–high-temperature metamorphic rocks, and Paleozoic passive-margin strata in its core (Pullen and Kapp, 2014; Xu et al., 2020). Jurassic shallow-marine strata are widespread along the limbs of the anticlinorium (Ma et al., 2023a). During Jurassic–Early Cretaceous time, the Bangong-Nujiang (Meso-Tethys) oceanic plate subducted northward beneath the Qiangtang terrane, resulting in continental arc magmatism with peak ages of 170–150 Ma and 110 Ma and an intervening magmatic lull at 140–130 Ma (Zhu et al., 2016). The Qiangtang terrane was uplifted above sea level during the latest Jurassic to Early Cretaceous due to Lhasa–Qiangtang suturing (Ma et al., 2017, 2018; Hu et al., 2022; Ma et al., 2023a). The nonmarine red beds of the Abushan Formation were deposited in intermontane basins during mid- to Late Cretaceous time, along with coeval (100–70 Ma) high-K volcanic rocks (Li et al., 2013b; He et al., 2018; Ma et al., 2023b; Wen et al., 2023), indicating that the Qiangtang terrane had become a significant orogenic belt prior to the Cenozoic India–Asia collision. Cenozoic red beds in the Qiangtang terrane have generally been assigned to the Kangtuo and Suonahu Formations without systematic geochronologic constraints (Mou et al., 2005; Yue et al., 2006). Distinguishing these formations from each other and from the Abushan Formation is challenging because of the lack of robust age control and similarity in lithology among these formations. In addition, the Qiangtang terrane exposes weakly deformed

to undeformed Eocene to Oligocene alkaline and adakitic volcanic rocks within scattered elliptical areas, among which the Dogai complex is the largest (Fig. 1B; Wang et al., 2008b; Kapp and DeCelles, 2019; Zeng et al., 2021; McMillan and Schoenbohm, 2023).

The Xianqian Basin is oval-shaped and located in the southwest part of the Qiangtang terrane (Figs. 1B and 2A). Most red beds within the Xianqian Basin were previously mapped as the Cretaceous Abushan Formation (1:250,000 geologic map of the Jiaco area by the Chinese Geological Survey; He et al., 2006). However, in this study, we demonstrated their Eocene depositional age through U–Pb zircon dating of interbedded tuffs and stratigraphic correlation. We thus define the Maerguo Formation as a distinct stratigraphic unit in the Xianqian Basin. The Maerguo Formation unconformably overlies or is structurally bounded by Lower Cretaceous felsic volcanic rocks, 165–160 Ma granitoid rocks, Paleozoic and Jurassic limestone and clastic rocks, and Cretaceous nonmarine red beds in different locations along the basin margins (Fig. 2A).

Several other Cenozoic nonmarine basins are scattered along the Bangong-Nujiang suture zone, ~30 km to the south of the Xianqian Basin, including the Lunpola–Nima, Gerze, and Awenco basins from east to west (Fig. 1B). The Bangong-Nujiang suture includes mainly Jurassic deep-marine turbidite-bearing strata and ophiolites that represent the relics of the subducted Bangong-Nujiang (Meso-Tethys) Ocean (Hu et al., 2022). This suture was uplifted above sea level during the Late Jurassic–Early Cretaceous collision between the Lhasa and Qiangtang terranes, which was diachronous along strike (decreasing in age from east to west; Kapp et al., 2007; DeCelles, et al., 2007; Ma et al., 2018, 2024). The Cenozoic basins postdate closure of the Meso-Tethys Ocean and provide important geologic records for postcollisional paleoenvironment and paleoaltimetry. We explore the development of these basins in relation to the Xianqian Basin later in the paper to better understand the regional development of topography and paleoenvironmental changes in central Tibet.

3. METHODS

3.1. Field Geology

Stratigraphic sections were measured using a tape measure at the decimeter scale. Lithofacies codes adapted from Miall (1977) and DeCelles et al. (2007) were used for logging and interpreting sedimentary features (Table 1). Paleocurrent data were measured from dune-scale trough

and planar cross-stratification (DeCelles et al., 1983). Samples for geochronology, sedimentary petrology, and paleontology were collected in the context of the measured sections. For conglomerates, the lithologies of ~100 clasts were determined within a 1 m² area in each outcrop site for provenance interpretation; the results are provided in Table S1 in the Supplemental Material.¹

3.2. Framework Composition and Heavy-Mineral Suite of Sandstone

In total, 73 sandstone samples from the Maerguo Formation were prepared as 30-μm-thick thin sections for petrographic analysis. The modal framework-grain composition was determined by point counting ≥400 grains per slide under a polarized microscope, following the Gazzi–Dickinson method (Gazzi, 1966; Ingersoll et al., 1984). Grain types and results are provided in Table S2.

Fifteen samples from the Maerguo Formation were selected for heavy-mineral analysis. Sandstone samples were hand-crushed using an agate mortar, and the 32–500 μm size fraction was obtained by wet sieving. Heavy minerals were separated by centrifugation in sodium polytungstate (density: 2.90 g/cm³) and recovered through partial freezing with liquid nitrogen. The dense fraction was then mounted on a slide using Canada balsam, and ≥200 transparent heavy minerals (or all present in the grain mount) were point-counted at regular intervals to determine accurate volume percentages (Garzanti and Andò, 2019). Transparent heavy-mineral assemblages were defined as the spectrum of detrital allochthonous minerals with a density >2.90 g/cm³, identifiable under a transmitted light microscope. The sum of zircon, tourmaline, and rutile relative to the total transparent heavy minerals (ZTR index of Hubert, 1962) quantifies the relative proportion of durable minerals within the assemblage and serves as an indicator of sediment recycling (Garzanti, 2017). These results are provided in Table S3.

3.3. Zircon U–Pb Dating

Zircons were extracted from seven sandstone (23NR84, 23NR18, 23GZ46, 23NR54, 23NR60, 23TG01, and 23TG14), four magmatic rock (23GZ41, 23NR83, 23NR50, and 23XQ271), and one tuff (23XQ316) samples through crush-

¹Supplemental Material. Figures and dataset supporting this study. Please visit <https://doi.org/10.1130/GSAB.S.30359200> to access the supplemental material; contact editing@geosociety.org with any questions.

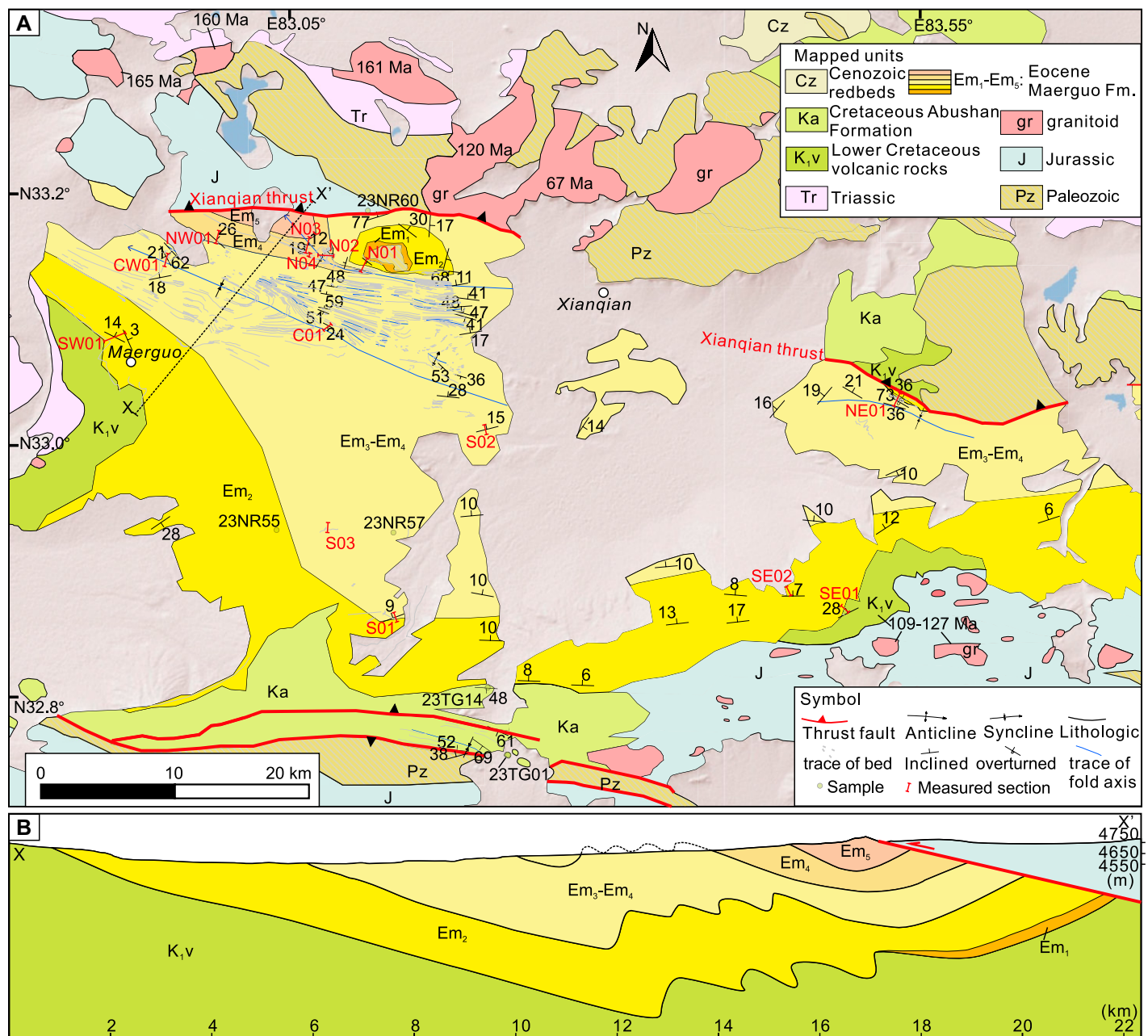


Figure 2. (A) Geologic map and (B) schematic geologic cross section of the Xianqian Basin. Geologic map was modified after the map produced by the Chinese Geological Survey (Mou et al., 2005) by integrating our own mapping data. Ages of igneous rocks are from the Tibetan Magmatism Database (Chapman and Kapp, 2017).

ing, elutriation, and magnetic separation. For detrital zircons from sandstone, ~120 grains were dated per sample; for magmatic rock and tuff, ~20 grains were analyzed per sample.

Zircon grains from the sandstone and magmatic rocks were randomly picked and mounted in epoxy and polished to expose grain interiors. Zircon U, Th, and Pb isotopes were measured at the State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits, Nanjing University, utilizing an ASI RESOLU-

tion S-155 193 nm ArF excimer laser coupled with a Thermo Scientific iCAP Qc quadrupole inductively coupled plasma-mass spectrometer (ICP-MS). U-Pb age data were collected by laser ablation with a beam diameter of 30 μm , a beam energy density of ~3.5 J/cm², and a repetition rate of 5 Hz. Each analysis included 20 s of background, 50 s of ablation, and 20 s of washout. With an ablation rate of ~0.1 μm /pulse, 50 s of ablation with a 30 μm beam diameter produces an ablation crater with a diameter/

depth ratio of <1. The $^{207}\text{Pb}/^{206}\text{Pb}$, $^{206}\text{Pb}/^{238}\text{U}$, $^{207}\text{Pb}/^{235}\text{U}$ ($^{235}\text{U} = ^{238}\text{U}/137.88$), and $^{208}\text{Pb}/^{232}\text{Th}$ ratios were corrected using the zircon standard 91500 as an external standard (isotope dilution-thermal ionization mass spectrometry $^{206}\text{Pb}/^{238}\text{U}$ age date at 1062.4 ± 0.8 Ma; Wiedenbeck et al., 1995). Plešovice and GEMOC GJ-1 standards were applied as quality-control references. Zircon standards were analyzed every 15 unknown samples. Data reduction was performed with ICPMSDataCal software (Liu et al., 2010). The

TABLE 1. LITHOFACIES FOR SEDIMENTARY ROCKS OF THE MAERGUO FORMATION IN THE XIANQIAN BASIN, WITH INTERPRETATIONS OF THEIR DEPOSITIONAL PROCESSES

Lithofacies code	Description	Interpretation
Gcm	Clast-supported, pebble-cobble, poorly sorted, unstratified	Deposition from clast-rich debris flows and sheetfloods
Gch, Gchi	Clast-supported, pebble-cobble, medium to well sorted, horizontally stratified, imbricated (long-axis transverse to paleoflow)	Shallow traction currents in longitudinal bars, or sieve deposits in alluvial fans
Gt	Clast-supported, pebble to boulder, trough cross-stratified conglomerate	Deposition by large gravelly ripples under traction flows in relatively deep, stable fluvial channels
Sm	Massive medium- to coarse-grained sandstone, sometimes pebbly and lens-like	High rate of deposition, highly concentrated, sand-rich deposits, probably formed during high-discharge periods
Sh	Medium- to very coarse-grained sandstone, horizontally stratified, sometimes pebbly	Plane-bed flow under strong or very shallow unidirectional flow
St	Fine- to coarse-grained sandstone, trough stratified	Sinuously crested and linguoid (3-D) dunes under moderately powerful, unidirectional flow in large channels
Sp	Medium- to very coarse-grained sandstone with planar cross-stratification	Migration of large two-dimensional (2-D) ripples under moderately powerful, unidirectional channelized flows or migration of sandy transverse bars
Sr	Fine- to medium-grained sandstone, with asymmetric two- and three-dimensional (3D) current ripples	Migration of small 2-D and 3-D ripples under weak unidirectional flows in shallow channels
Sb	Bioturbated sandstone	Trace fossils formed by insects, dwelling, resting, crawling
F	Siltstone and mudrock	Suspension setting in overbank, abandoned channel, waning flood deposits, or subaqueous lake
M	Marlstone/micritic carbonate	Lacustrine carbonate mud with insignificant siliciclastic influx
Gyp	Gypsum	Lacustrine evaporite in an arid environment
Vb	Tuffaceous breccia, thick- to medium-bedded, gray, including breccia of pebbles and granule size and dominated by mudrock	Pyroclastic flow into the lake
Vt	Volcanic glass-dominated tuff, mostly thin to medium bedded, with massive-bedded locally, and gray or green in color	Suspension fallout, or transported though river punching into the lake

Note: Modified after Miall (1977) and DeCelles et al. (2007).

measured Plešovice and GJ-1 yielded weighted mean $^{206}\text{Pb}/^{238}\text{U}$ ages of 337.2 ± 2.0 Ma ($n = 23$, mean square of weighted deviates [MSWD] = 0.09) and 599.8 ± 2.0 Ma ($n = 55$, MSWD = 0.038), respectively (Fig. S2), in agreement with the recommended $^{206}\text{Pb}/^{238}\text{U}$ ages of Plešovice ($^{206}\text{Pb}/^{238}\text{U}$ age = 337 Ma; Sláma et al., 2008) and GJ-1 ($^{206}\text{Pb}/^{238}\text{U}$ age = 602.7–598.5 Ma; Jackson et al., 2004). The $^{206}\text{Pb}/^{238}\text{U}$ zircon age was used for interpretation where the $^{206}\text{Pb}/^{238}\text{U}$ age was younger than 1000 Ma, and the $^{207}\text{Pb}/^{206}\text{Pb}$ age was used if the $^{206}\text{Pb}/^{238}\text{U}$ age was older than 1000 Ma. Ages older than 400 Ma with >30% ($^{207}\text{Pb}/^{235}\text{U}$ vs. $^{206}\text{Pb}/^{207}\text{Pb}$) discordance were removed from consideration, while grains younger than 400 Ma with >10% discordance ($^{207}\text{Pb}/^{235}\text{U}$ vs. $^{206}\text{Pb}/^{238}\text{U}$) were also excluded. The results are provided in Table S4.

Zircons from tuff sample 23XQ316 and standards (Plešovice and Qinghu) were mounted in a single epoxy target and polished to expose grain interiors. Secondary ion mass spectrometry (SIMS) U-Pb dating was conducted using a Cameca IMS-1300HR³ at the State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits, Nanjing University. Reflected and transmitted light microscopy and cathodoluminescence (CL) imagery were used to aid the selection of analysis spots. A 3 nA primary O^- ion beam spot $\sim 15 \times 10$ μm in size was applied to impact on the gold-coated surface of zircon grains. Positive secondary ions were extracted with a 10 kV potential. In the secondary ion beam optics, a 60 eV energy window was used, together with a mass resolution of ~ 7000 ($M/\Delta M$; at 50% peak height), to

separate Pb^+ peaks from isobaric interferences. A single electron multiplier was used in ion-counting mode to measure secondary ion beam intensities by peak-jumping mode. Each measurement consisted of eight cycles. Analyses of the standard zircon Plešovice were interspersed with unknown grains. Pb/U calibration was performed relative to the zircon standard Plešovice ($^{206}\text{Pb}/^{238}\text{U}$ age = 337 Ma; Sláma et al., 2008). Measured compositions were corrected for common Pb using nonradiogenic ^{204}Pb . Corrections were sufficiently small to be insensitive to the choice of common Pb composition, and an average of present-day crustal composition (Stacey and Kramers, 1975) was used for the common Pb, assuming that the common Pb was largely surface contamination introduced during sample preparation. In order to monitor the external uncertainties of SIMS U-Pb zircon dating calibrated against the Plešovice standard (Sláma et al., 2008), an in-house zircon standard Qinghu (Li et al., 2013a) was alternately analyzed as an unknown. The isotopic and age dataset obtained by SIMS is detailed in Table S5. The U-Pb concordia diagrams and corresponding weighted mean $^{206}\text{Pb}/^{238}\text{U}$ age plots for the zircon standards are shown in Figure S2.

3.4. Carbonate U-Pb Dating

Carbonate U-Pb dating of one limestone sample (23NR93) was carried out using a Thermo Fisher Neptune Plus multicollector (MC) ICP-MS in combination with a RESolution S155LR 193 nm ArF excimer laser-ablation system at the Mineral Laser Microprobe Analysis Laboratory (Milma Lab), China University of Geosciences,

Beijing (CUGB). The detailed instrumental and experimental parameters were described in Zhang et al. (2021). Data were acquired on thin sections. Due to the widespread element and isotope heterogeneity of carbonate minerals, the grid diagram method was used to divide the sample into many regions, and then each region was ablated sequentially under optimized instruments parameters with ^{238}U signal intensity of 8–12 mV and $^{207}\text{Pb}/^{206}\text{Pb}$ of Standard Reference Material (SRM) NIST-614 of ~ 1 (Zhang et al., 2021). Finally, the areas with relatively high U (^{238}U signal intensity >2.5 mV) and low common Pb content (variable or low $^{207}\text{Pb}/^{206}\text{Pb}$ ratios) were selected as the formal ablation areas. A laser spot size of 90 μm , a laser energy density of 2 J/cm², and a repetition rate of 10 Hz were employed during the analysis. Helium was used as carrier gas, while argon and nitrogen were used as makeup and enhancement gases. Each analysis was composed of 20 s background time followed by 30 s of ablation and then 30 s of washout. Unknown samples were corrected by standard bracketing with synthetic SRM NIST-614 glass for instrumental drift and Pb isotope composition (Woodhead and Hergt, 2000) and a natural calcite WC-1 of 254.4 ± 6.4 Ma (Roberts et al., 2017) to correct for $^{206}\text{Pb}/^{238}\text{U}$ fractionation. Calcite reference material Tarim with an age of $208.0 \pm 0.4/3.2$ Ma (Zhang et al., 2023) was measured to provide a control on accuracy and precision. These measurements yielded Tera-Wasserburg concordia intercept ages of 207.8 ± 1.4 Ma ($n = 22$, MSWD = 0.38), which is consistent with the published ages within the analytical uncertainty. Raw data reduction was performed off-line using Iolite

v3.73 software (Paton et al., 2011). The IsoPlot 4.15 software (Ludwig, 2008) was used to calculate the U-Pb ages and generate the Tera-Wasserburg diagrams. Age errors are presented at the 95% confidence level, and error ellipses are given as 2σ . The data results and Tera-Wasserburg diagrams of calcite reference material Tarim are provided in Table S6.

3.5. Tectonic Subsidence Analysis

The tectonic subsidence analysis was based on the high-precision age-depth data from the Xianqian Basin, following the backstripping method outlined by Allen and Allen (2005) to remove the effects of sediment loads and water-depth variations. Decompaction of stratigraphic units was applied to eliminate the effect of the exponential decrease of porosity with depth during progressive burial, before an Airy isostatic model was adopted to subtract sediment load, with the sediment lithologies and densities determined based on our measured section and comparable litholo-

gies listed in Allen and Allen (2005). Paleowater depth deepened from 0 m in the alluvial and fluvial units Em_1 and Em_2 to >5–15 m in the lacustrine unit Em_4 , which contains gravity-flow deposits, based on our sedimentologic observations and interpretations (see next section).

4. RESULTS

4.1. Stratigraphy and Sedimentology

The Eocene Maerguo Formation consists of an ~930 m fining-upward subsequence and then an ~967 m coarsening-upward subsequence, which combined reach nearly 2 km in thickness (Fig. 3). Preliminary mapping showed that the Maerguo Formation is strongly deformed, with tight folds in the central Xianqian Basin that broaden to the north and south (Fig. 2B). In the northern Xianqian Basin, the Maerguo Formation unconformably overlies Mesozoic sedimentary and granitoid rocks (Fig. 4A). The basal unconformity is overlain by the alluvial lower conglomerates (unit

Em_1), which transitions laterally and upward into the fluvial red beds (unit Em_2). Unit Em_2 gradually transitions into the lower fine-grained lacustrine red beds (unit Em_3). Further up section, the upper lacustrine unit (unit Em_4) includes facies (Figs. 4B and 4H) indicative of gravity-flow deposition in a deeper lacustrine environment than unit Em_3 , before finally being capped by alluvial upper conglomerates (unit Em_5 ; see Fig. 4C). The northern margin of the Xianqian Basin is bounded by the north-dipping Xianqian thrust fault, which places pre-Cretaceous granitoid, limestone, and clastic rocks in the north (Figs. 4C and 4D) and Cretaceous volcanic rocks in the northeast (Fig. 4E) in the hanging wall over Eocene red beds of the Maerguo Formation in the footwall. Growth strata are locally present within the Maerguo Formation, suggesting that deposition was contemporaneous with motion on the Xianqian thrust (Fig. 4E).

In the southern Xianqian Basin, the Maerguo Formation unconformably overlies Cretaceous volcanic rocks and Mesozoic sedimentary strata

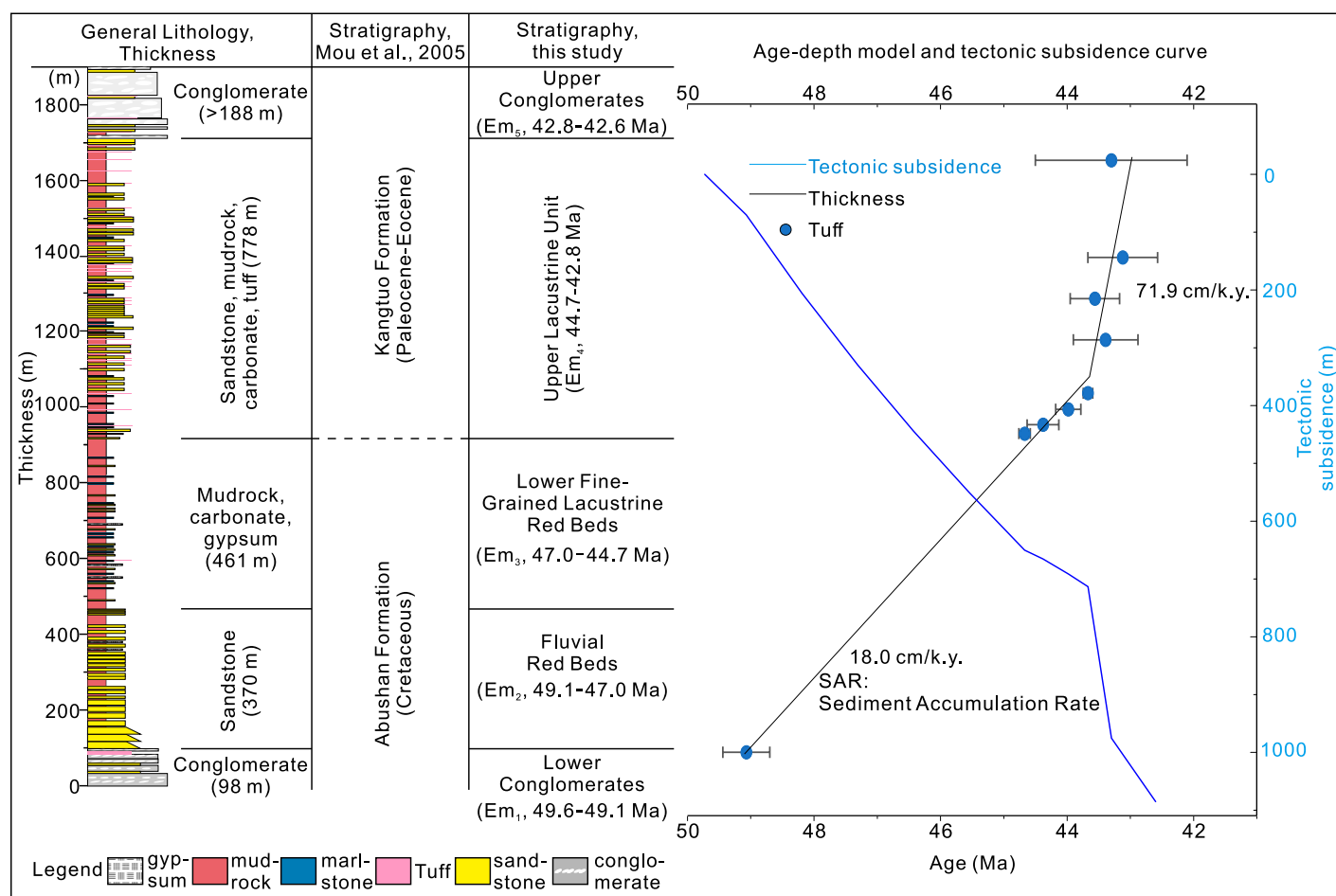


Figure 3. Integrated stratigraphic chart for the Xianqian Basin. The former stratigraphic scheme of the Chinese Geological Survey (Mou et al., 2005) is shown for comparison. The age-depth model and tectonic subsidence curve were plotted based on tuff age points reported in A. Ma (2025, personal comm.).

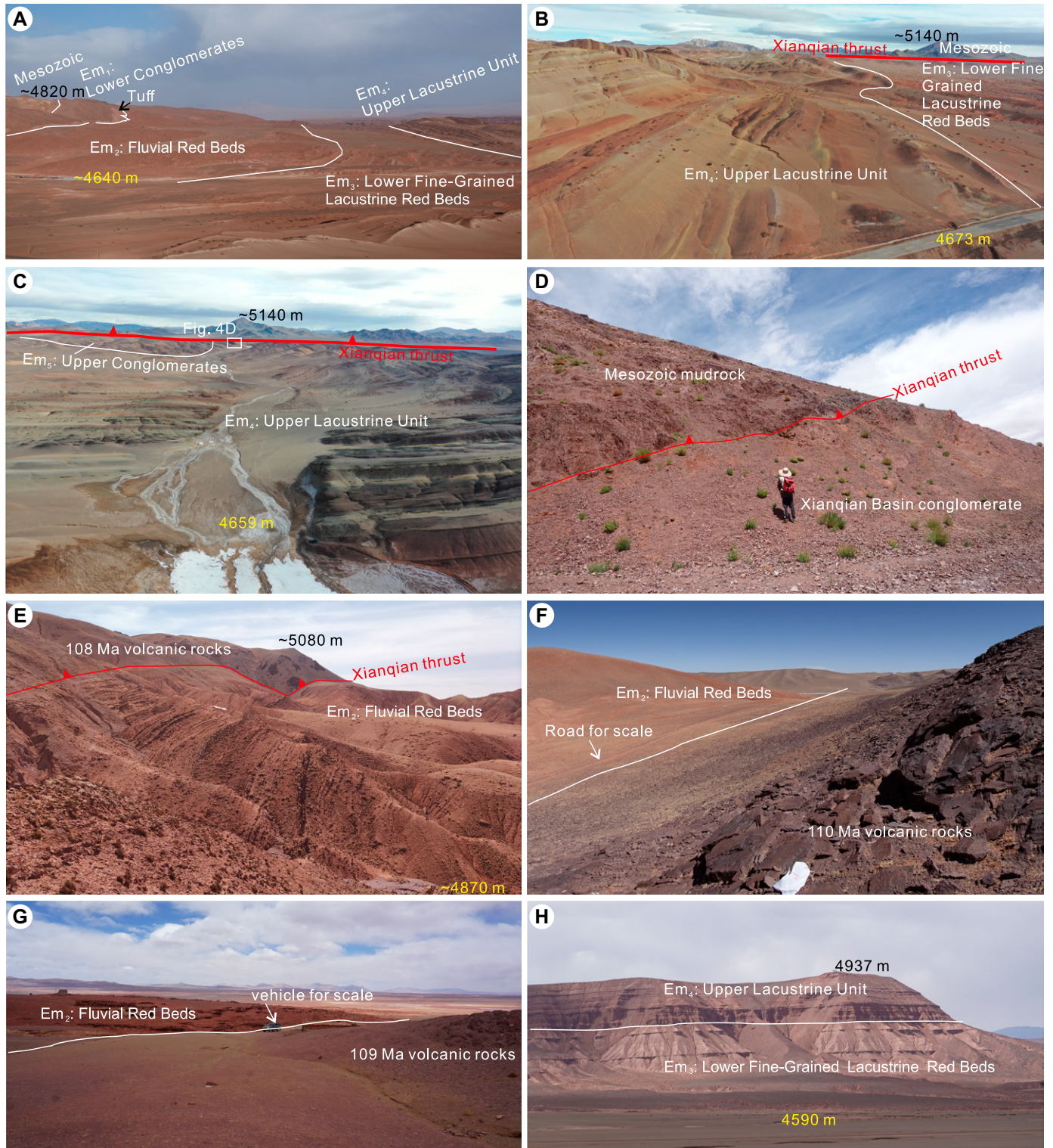


Figure 4. Field photos showing major geologic contacts. (A) Bottom of the Xianqian Basin strata, section N01. View toward the west. (B) Stratigraphic transition from unit Em₃ to unit Em₄, section N02. View toward the north. (C) Top of Xianqian Basin strata, sections N03 and N04. View toward the north. (D) North-dipping Xianqian thrust juxtaposing Mesozoic mudrock in the hanging wall against the upper lacustrine unit and upper conglomerates in the footwall. View toward the northeast. (E) Circa 108 Ma volcanic rocks thrust over the Xianqian Basin, synchronous with sedimentation of unit Em₂ in the footwall, which show growth strata with overturned beds in the proximal footwall, section NE01. View toward the east. (F) Unit Em₂ unconformably overlying ca. 110 Ma volcanic rocks, section SE01. View toward the west. (G) Unit Em₂ unconformably overlying ca. 109 Ma volcanic rocks, section SW01. View toward the north. (H) Stratigraphic transition from unit Em₃ to unit Em₄, section S01. The cliff is ~350 m high. View toward the north.

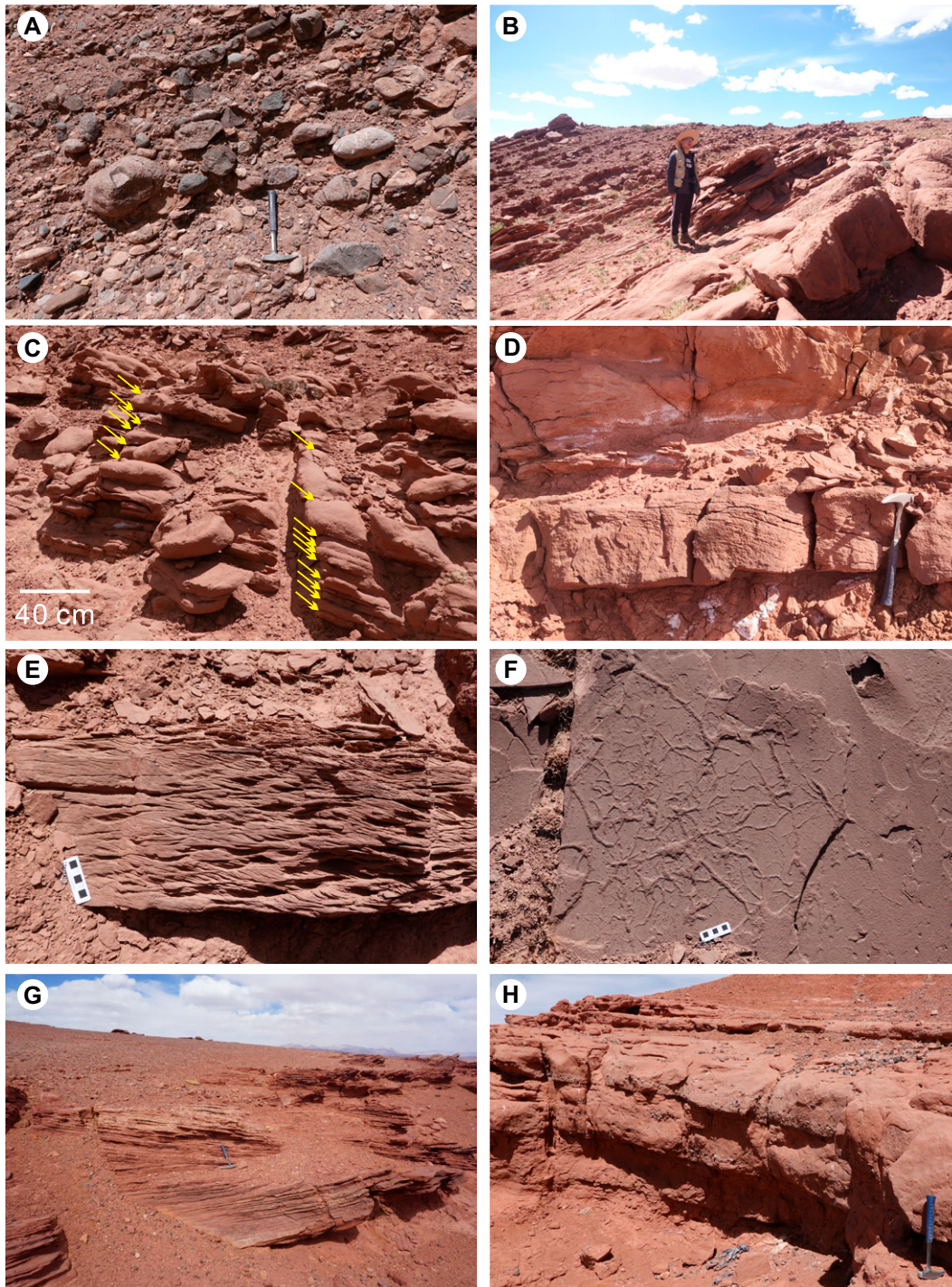


Figure 5. Field photos showing the typical lithofacies in units Em₁, Em₂, and Em₃. (A) Crudely horizontally stratified and clast-imbricated conglomerate (Gchi), unit Em₁, section N01. (B) Alternation of red sandstone (Sh, St, Sm, Sb, Sp, and Sr) and mudrock (F), unit Em₂, section N01. (C) Sandstone with horizontal stratification (Sh) is organized into multiple beds bounded by finer-grained materials, as indicated by yellow arrows, unit Em₂, section N01. (D) Trough cross-bedded sandstone (St) in the overturned unit Em₂, section NE01. (E) Fine-grained sandstone with climbing ripples (Sr), unit Em₂, section N02. (F) Bioturbated sandstone (Sb) with trace fossil on top, unit Em₂, section N01. (G) Planar cross-bedded sandstone in unit Em₂, section SW01. (H) Multi-story upward-fining sandstone bodies bounded by pebbles at the base, indicative of channel lag deposits in braided rivers, unit Em₂, section SE02.

(Figs. 4F and 4G). Unit Em₁ is absent; unit Em₂, characterized by red sandstone and mudrock, directly overlies the Cretaceous volcanic rocks or Jurassic sedimentary rocks. The contacts between the units within the Maerguo Formation in the southern basin are conformable (Fig. 4H). Unit Em₃ also does not crop out in the southern Xianqian Basin.

Below, we describe the sedimentologic features of each stratigraphic unit. The representa-

tive lithofacies photos are provided in Figures 5 and 6, and logs of measured sections are provided in Figures 7–10.

4.1.1. Unit Em₁: Lower Conglomerates

4.1.1.1. Description. Unit Em₁ (lower conglomerates) is 98 m thick in section N01 in the northern Xianqian Basin. This unit consists of an overall fining-upward sequence of boulder to pebble conglomerate with minor sandstone

interbeds. In the lower part of section N01, the conglomerates are poorly sorted, with maximum clast sizes reaching ~35 cm along the long axis. Clast size decreases to ~15 cm and sorting improves to moderate in the upper part of the section. The conglomerate is clast-supported, poorly to moderately sorted, and primarily displays crude horizontal stratification (lithofacies code Gch). It commonly exhibits imbrication with long axes oriented transverse to the paleo-

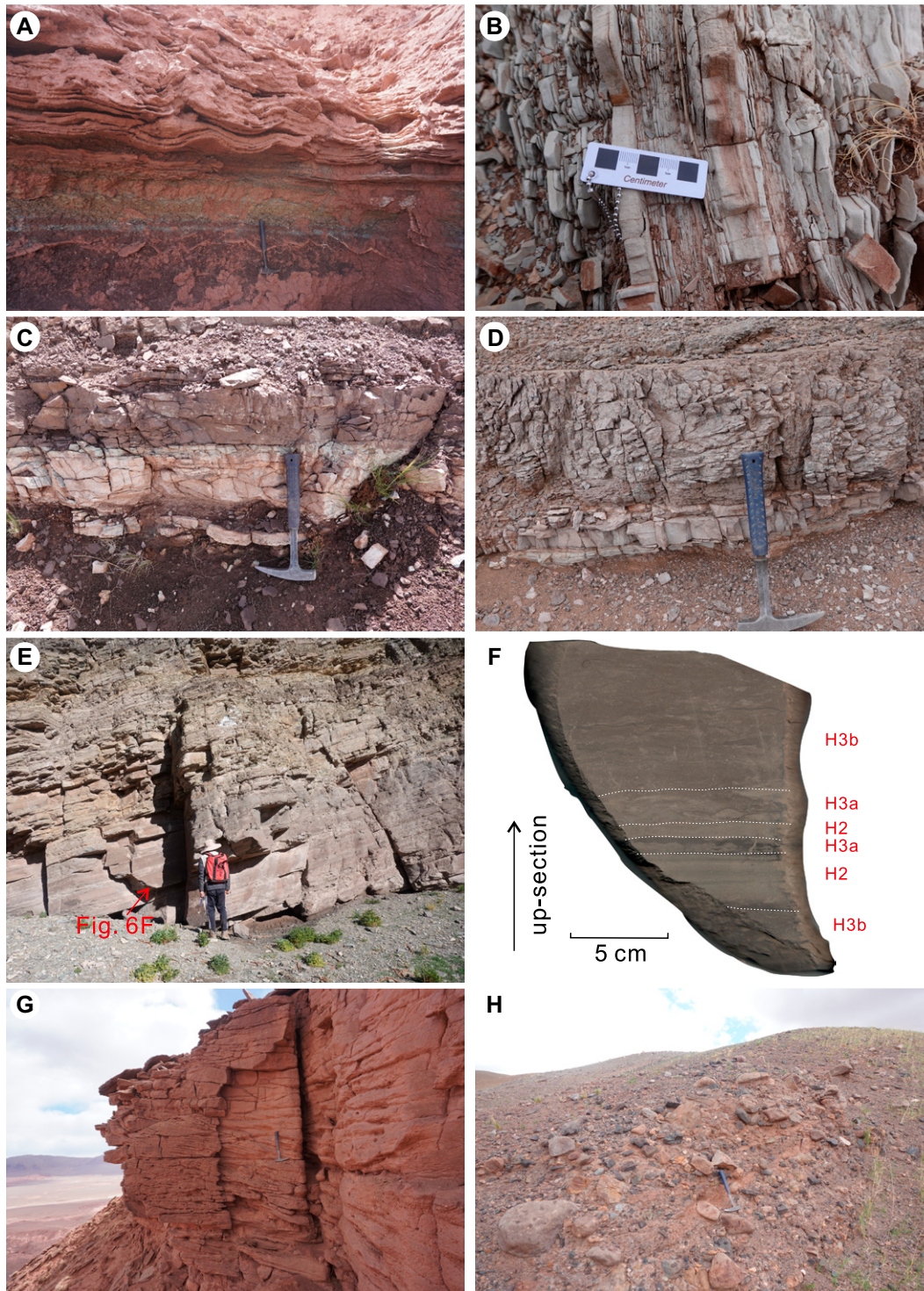


Figure 6. Field photos showing the typical lithofacies in units Em₃, Em₄, and Em₅. (A) Gypsum (Gyp) with structurally deformed laminations, along with green micrite (M) intralayers, unit Em₃, section S01. (B) Thinly interbedded micritic carbonate (M), unit Em₃, section N01. (C) White-colored tuff (Vt) beneath the sandstone, unit Em₄, section N02. (D) Medium-bedded clast-bearing tuffaceous breccia (Vb) overlying the thinly bedded tuff (Vt), unit Em₄, section N04. (E) Thickly to thinly bedded tabular gray layers of sandstone and mudrock with good lateral continuity, deposited through sublacustrine gravity flows, unit Em₄, section N03. (F) Hybrid event beds with three internal divisions in a sample collected in beds as shown in part E: H2—alternating lighter and darker sands; H3—mud-rich debritic sandstone, where subunit H3b contains more mud than H3a. Interpretation is based on Haughton et al. (2009) and Dodd et al. (2022). (G) Superposition of trough cross-bedded sandstone (St), unit Em₄, section S01. (H) Clast-supported massive conglomerate (Gcm), unit Em₅, section N03.

flow direction (Gchi; Fig. 5A), while cross-stratified conglomerates (Gt) are relatively rare. The conglomerates are arranged in 1–2-m-thick beds, typically with erosional basal surfaces and finer-grained upper parts. Lenticular beds of massive granular to coarse-grained sandstone (Sm) are locally intercalated. These lithofacies are grouped within (litho)facies association 1 (FA 1).

4.1.1.2. Interpretation. The relatively scarce occurrence of gravity-flow lithofacies (e.g., disorganized Gmm and Gcm) and the abundance of organized conglomerate (Gch, Gchi, and Gt) suggest that most deposition took place in stream flows, likely during flash floods. The spatially localized occurrence of this unit, along with its lateral transition into amalgamated fluvial sand-

stones, indicates deposition within stream-dominated alluvial fans (Ori, 1982; Wilford et al., 2005; DeCelles et al., 2011). These strata are less likely to have been deposited in more distal braided rivers on the alluvial plain, as the conglomerate clasts are generally poorly sorted, and large-scale cross-stratified sandstone is absent. Based on the southward sediment transport

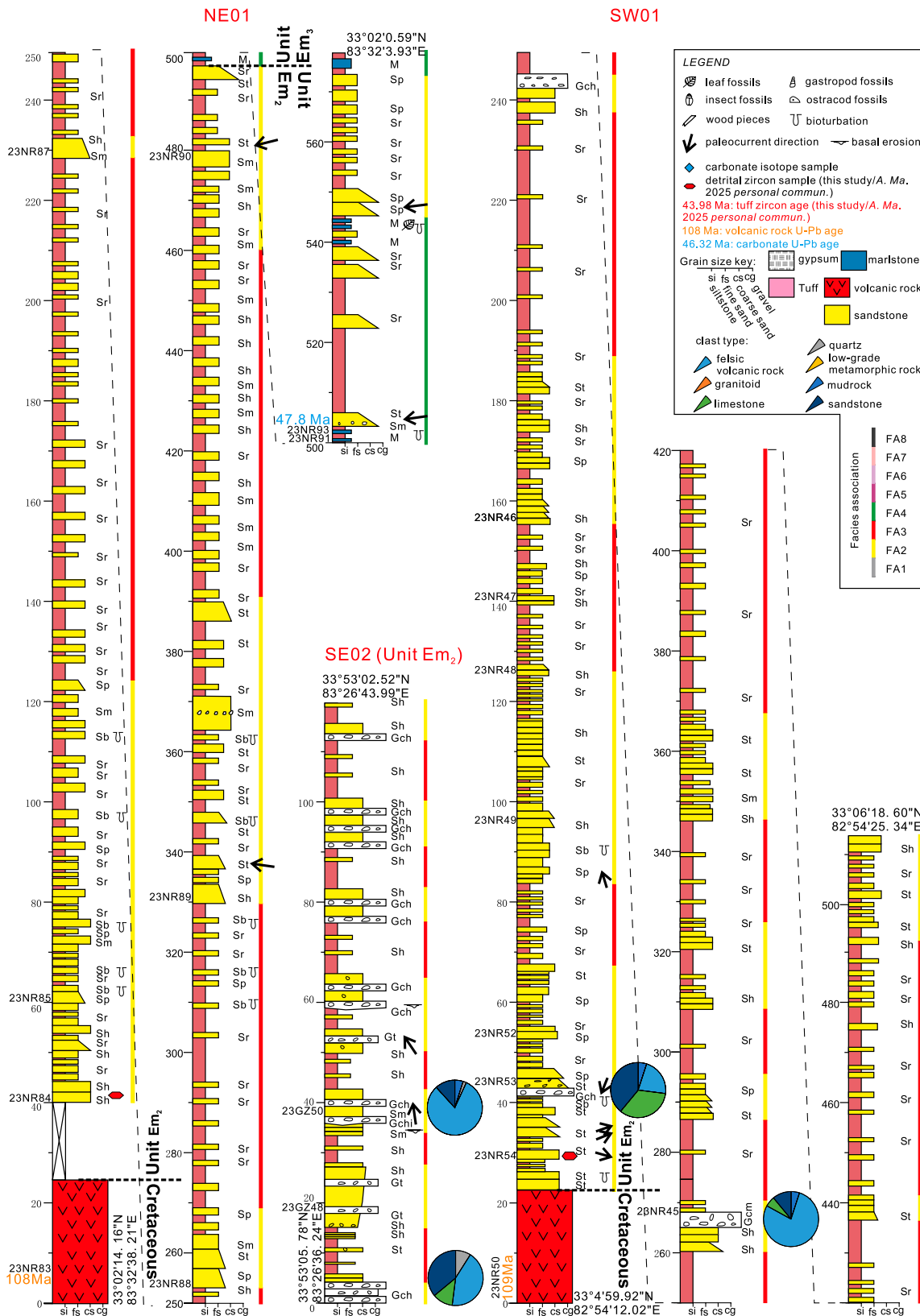


Figure 7. Logs of measured sections NE01, SW01, and SE02. Lithofacies codes refer to Table 1. FA 1—stream-dominated alluvial-fan deposits; FA 2—river channel and levee deposits; FA 3—overbank deposits; FA 4—shallow, muddy playa deposits; FA 5—river channel deposits on delta plain; FA 6—sublacustrine gravity-flow deposits in a delta front to prodelta; FA 7—river channel deposits on delta front; FA 8—gravity flow-dominated alluvial-fan deposits.

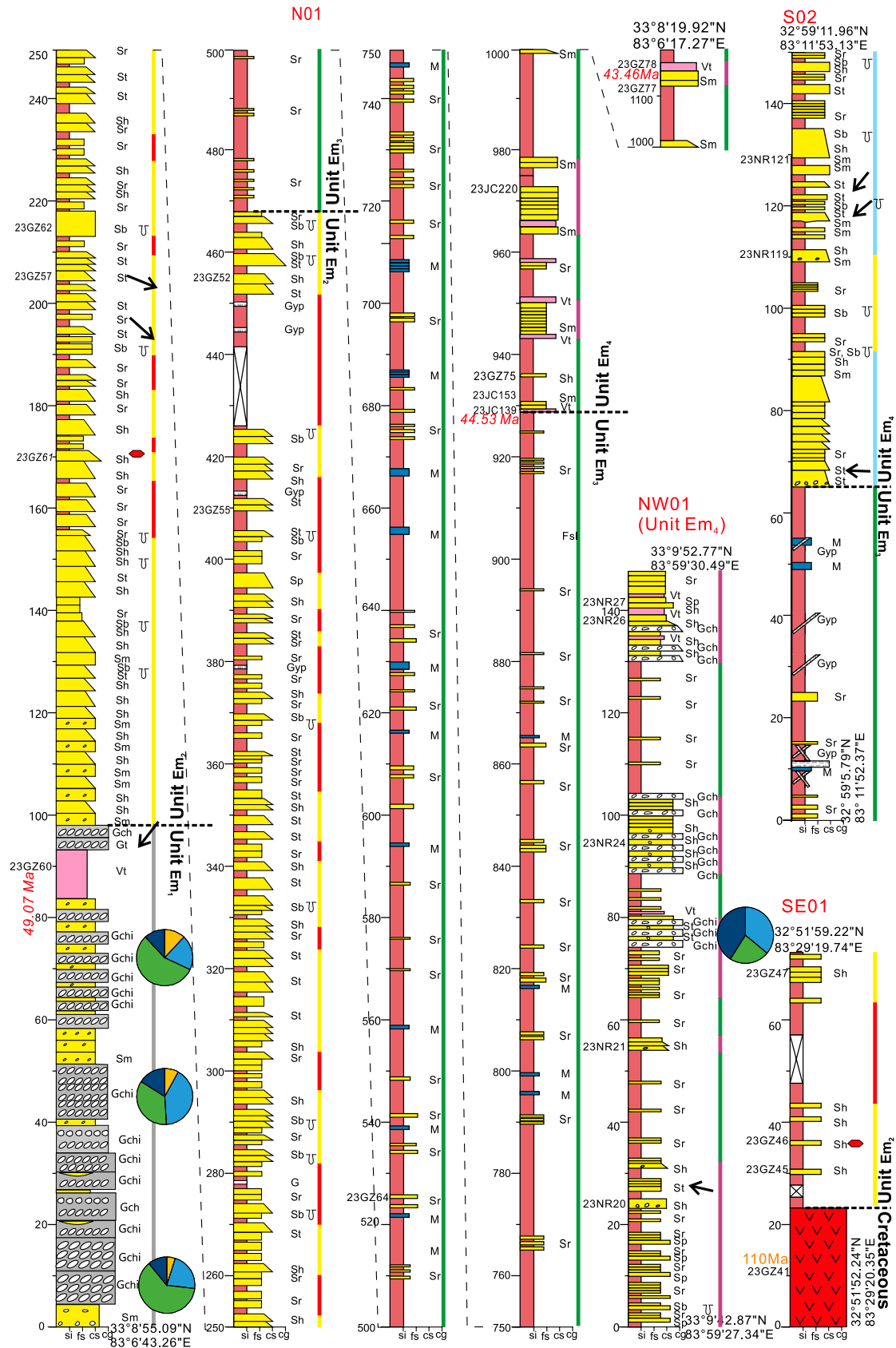


Figure 8. Logs of measured sections N01, S02, SE01, and NW01. Lithofacies codes refer to Table 1. See Figure 7 for legend.

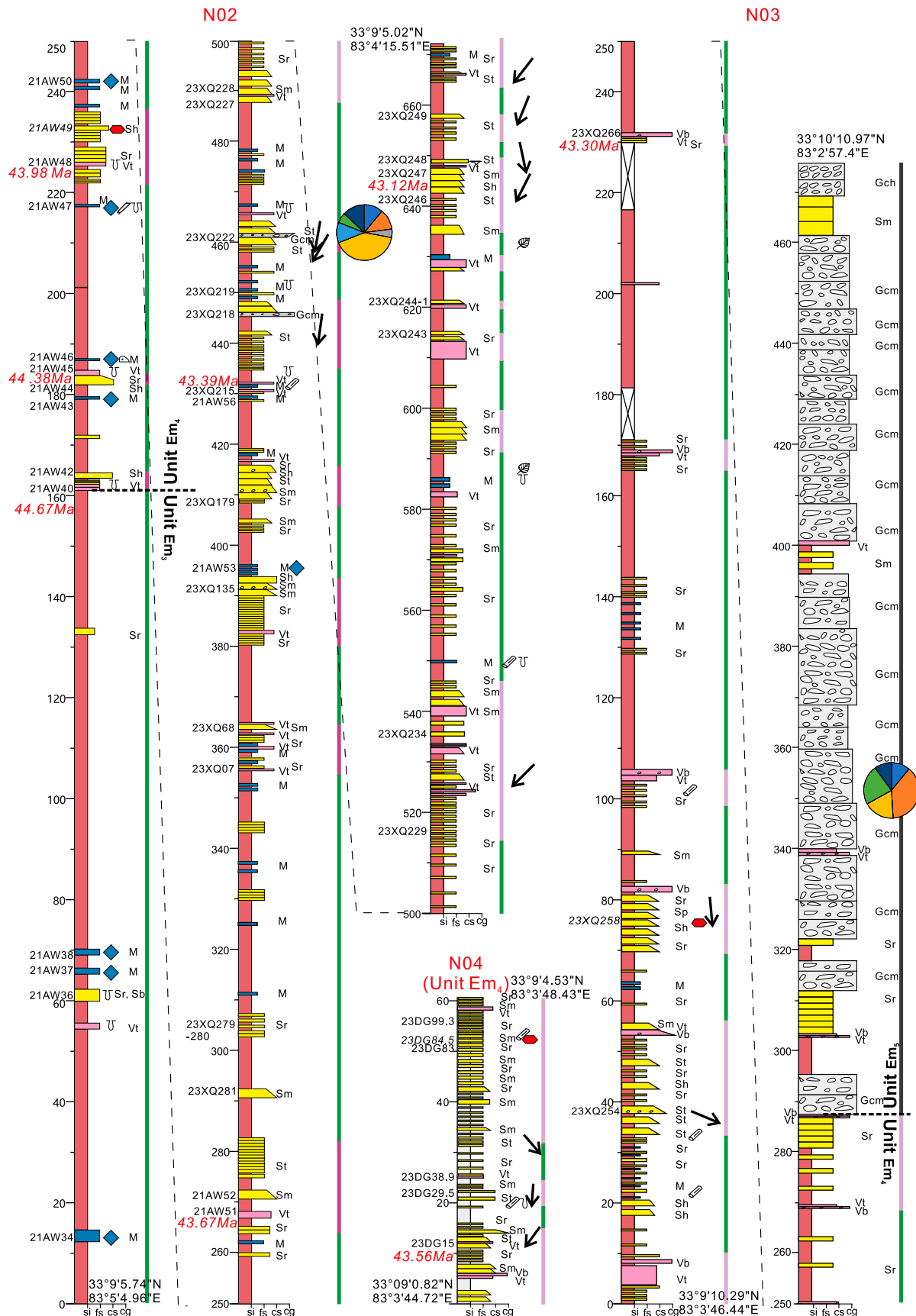


Figure 9. Logs of measured sections N02, N03, and N04. Lithofacies codes refer to Table 1. See Figure 7 for legend.



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inferred from conglomerate clast composition (Table S1), FA 1 in unit Em₁, exposed near the northern margin of the basin, aligns with deposition in alluvial fans that extended southward from high, rugged topography composed of limestone, felsic volcanic rock, sandstone, and low-grade metamorphic rocks within the Qiangtang terrane.

4.1.2 Unit Em₂: Fluvial Red Beds

4.1.2.1. Description. Unit Em₂ (fluvial red beds) is ~370 m thick in section N01, >455 m in section NE01, >50 m in section SE01, >120 m in section SE02, and >510 m in section SW01 (Figs. 7–8). This unit unconformably overlies the late Early Cretaceous volcanic rocks in the sections SE01 and SW01 (Figs. 4F–4G). In section N01, unit Em₂ contains sporadically distributed gypsum fragments within the mudrock at the top and conformably changes into unit Em₃. In contrast, the red beds of unit Em₂ in section NE01 are truncated in the footwall of the Xianqian thrust fault (Fig. 4E). In both sections N01 and NE01, the top of unit Em₂ is marked by a transgressive surface characterized by lacustrine carbonates. In addition, unit Em₂ in section NE01 is characterized by growth strata, with overturned coarse-grained sediments at the bottom and upright fine-grained sediments at the top (Fig. 4E).

Unit Em₂ consists of an upward-fining sequence of alternating red sandstone and mudrock (Figs. 5B) and includes two lithofacies associations (FA 2 and FA 3). FA 2 in unit Em₂ is dominated by medium- to thin-bedded sandstone with horizontal stratification (Sh; Fig. 5C), planar (Sp) and trough cross-stratification (St; Fig. 5D), ripples and climbing ripples (Sr; Fig. 5E), and bioturbation (Sb; Fig. 5F), along with minor mudrock (F). In the northern Xianqian basin, the sandstone in FA 2 is organized into multiple beds bounded by mudrock, as seen in sections N01 and NE01 (Fig. 5C). In the southern Xianqian Basin, sandstone with larger-scale planar cross-stratification (Fig. 5G) in FA 2 mainly occurs as multistory bodies that are bounded by pebble conglomerate, as found in sections SE02 and SW01. The conglomerate is horizontally stratified (Gch) and occasionally imbricated (Gchi) (Fig. 5H) with trough stratification (Gt). Erosive bases can be found at the bottom of some sandstone and conglomerate layers. Overall, sandstone and conglomerate layers in FA 2 are commonly lenticular with river channel geometry along strike. Sandstone and conglomerate in FA 2 become thinner and less common up section. Trace fossil burrows are found both perpendicular and parallel to sandstone layers (Fig. 5F). FA 3 is interbedded with FA 2 and includes mainly mudrock (F) and

minor Sr and gypsum fragments (Gyp). FA 3 occurs more common than FA 2 up section.

Paleoflow directions were determined by averaging at least five measurements of dune-scale trough cross-stratification at each site. The results suggest southward paleoflow for unit Em₂ in the sections N01 and NE01, and northward paleoflow in the southern sections SW01 and SE02.

4.1.2.2. Interpretation. Unit Em₂ is interpreted to have been deposited in a fluvial setting, including channel, levee, bars, and overbank deposits. Distinguishing between sand-dominated meandering and braided rivers based solely on outcrop evidence can be challenging, particularly when well-exposed point-bar lateral accretion surfaces, which are characteristic of meandering rivers and oriented perpendicular to the paleoflow direction, are absent (Gibling, 2006; Lyster et al., 2022; Wang et al., 2024).

FA 2 with lithofacies Sh, St, Sp, Gch(i), and Gt is common in fluvial channels. Sh and Gch(i) develop as upper-flow-regime plane beds under very shallow flows; St, Sp, Gt form by the migration of subaqueous sand dunes under unidirectional currents (e.g., Cant and Walker, 1978). Interfaces between multiple 5–20-cm-thick sand beds, characterized by finer-grained materials (Fig. 5C), could be interpreted as lateral accretion surfaces, suggestive of point bars in meandering river systems. However, these fine-grained materials may also occur in bars and overbanks of sandy braided rivers, which would be difficult to differentiate from laterally accreting point bars without detailed paleocurrent work (Miall, 1977; Wang et al., 2024). Multistory upward-fining sandstone bodies bounded by pebbles at their erosive bases (Fig. 5H) are indicative of channel lag deposits in braided rivers. Lithofacies Sr with climbing ripples in FA 2 indicates rapid sediment accretion associated with crevasse-splay deposition or bar-top deposits during waning floods. Compared to FA 2, the increasing appearance of lithofacies Sr and F in FA 3 is associated with overbank deposits. The occurrence of sporadic gypsum fragments within mudrock of FA 3 indicates deposition in arid environments. Unit Em₂ deposits were heavily bioturbated, as suggested by abundant trace fossils.

4.1.3. Unit Em₃: Lower Fine-Grained Lacustrine Red Beds

4.1.3.1. Description. Unit Em₃ (lower fine-grained lacustrine red beds) is ~461 m thick in section N01, >80 m thick in section NE01, >310 m in section S01, >65 m in section S02, >30 m in section S03, ~290 m in section CW01, and >25 m in section C01. This unit includes lithofacies association FA 4, which is

dominantly composed of red mudrock, subordinate sandstone, and minor light-gray limestone and gypsum (Figs. 6A and 6B). Sandstone beds are broadly tabular, decimeters to centimeters in thickness, generally fine-grained to very fine-grained (except for a few medium-grained sandstone beds near the tops of the sections), and notable for their abundant climbing ripple cross-stratification. Abundant centimeters-thick gypsum layers and net-like veins are present in the central and southern Xianqian Basin, as seen in sections S01, S02, S03, and CW01, but absent in the northern Xianqian Basin. Overall, the sandstone and gypsum layers are laterally continuous. The sandstones are well laminated, commonly topped by asymmetric ripples (Fig. 5H), and bioturbated in some horizons. The carbonate beds are centimeters in thickness. Plant fragments and gastropods (Fig. 11A) are found in the mudrocks. A few beds of tuff (among which sample 23NR56 from section S03 and sample 23XQ316 from section C01 were dated) are present within unit Em₃.

4.1.3.2. Interpretation. FA 4 in unit Em₃ is interpreted to have been deposited in a shallow, muddy playa environment. Sandstone deposition may have been related to quasi-steady or pulsed discharge of a sandy fluvial system entering an ephemeral lake, whereas the carbonate and gypsum likely formed during periods with insignificant river runoff. The gypsum vein was likely related to postdepositional disruption of original beds. The gastropod *Planorbidae* generally lives within calm water bodies and supports a lake interpretation (Strong et al., 2008). The presence of gypsum suggests a saline environment, especially in the central and southern Xianqian Basin, where gypsum deposition prevailed.

4.1.4. Unit Em₄: Upper Lacustrine Unit

4.1.4.1. Description. Unit Em₄ (upper lacustrine unit) is best exposed in the northern part of the basin, with a thickness of 778 m based on sections N01, N02, N03, N04, CW01, and NW01. Apart from FA 4, which also occurs in unit Em₃ and mainly consists of reddish and gray mudrock (F) with micritic carbonate (M) interlayers throughout unit Em₄, three other lithofacies associations (FA 5–7) were identified. FA 5 in the northern Xianqian Basin includes Sh, Sp, St, Sm, tuff (Vt), Gch(i), and clast-supported massive conglomerate (Gcm). The sandstone and conglomerate are typically restricted to lenticular bodies with erosional bases. The tuffs are mainly white, pink, and reddish in color and dominated by sand-sized volcanic ash grains (Fig. 6C). The sandstone and tuff are locally burrowed. FA 5 is found alternating with FA 4, forming thickening- and coarsening-upward

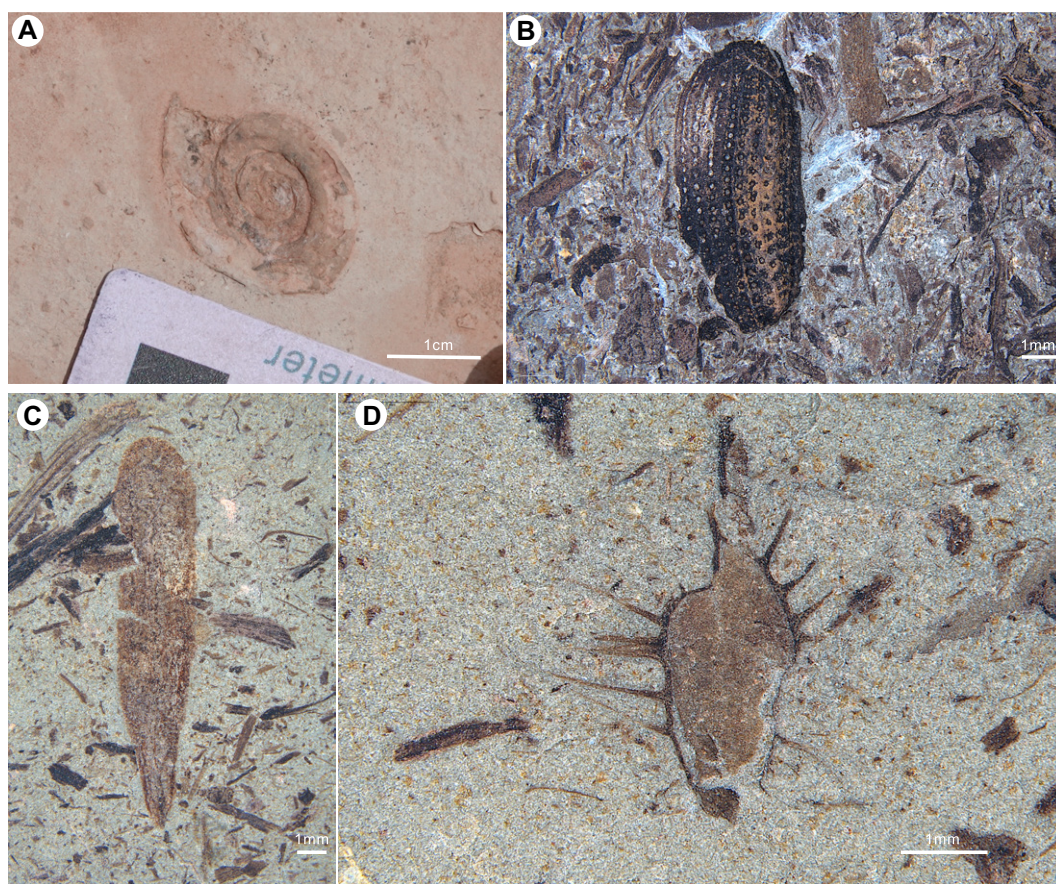


Figure 11. Examples of diverse fossil types and their stratigraphic horizons. (A) Gastropoda: Planorbidae, unit Em₃, section C01. (B) Insect fossil of Coleoptera, unit Em₄, section N02. (C) Representative leaf morphotype with narrow and elongated shape, unit Em₄, section N02. (D) Ceratophyllaceae Gray *Ceratophyllum* sp.? unit Em₄, section N02.

packages (up to several meters thick) in the basal 306 m of unit Em₄, as seen in sections N01–N03.

FA 6, characterized by tabular beds of sandstone, tuff, and tuffaceous breccia, punctuates the upper 472 m of unit Em₄ in the northern basin. These beds are laterally continuous at the kilometer scale. Tuffs (Vt) are similar in color and composition to those in the lower unit Em₄. Tuffaceous breccias (Vb) are pink to gray in color and contain mudrock, metamorphic rock, and granitoid clast (Fig. 6D). The sandstones are red to gray in color with lithofacies St, Sp, Sh, Sm, and Sr. Thickly to thinly bedded tabular gray layers of sandstone and mudrock, exhibiting good lateral continuity and upward-fining sequences, are stacked (Fig. 6E). An ideal individual sequence is typically characterized by a progression from clean, structureless, fine- to coarse-grained sandstone (Sm), “banding” of clean sandstone and light-gray mud-rich sandstone (Sm and F), mud-rich structureless sandstone with elongate mud clasts (Sm), to sandstone and mudrock with parallel and ripple laminations, and laminated mudrock (Sr and F) from bottom to top (Fig. 6F), though some lithofacies may be absent in a single bed. Alternating with FA 6, thin to very thinly bedded mudrock (F) and micritic carbonate (M) of FA

4 contain abundant fossilized insects of Coleoptera (Fig. 11B) and plant fragments (Figs. 11C and 11D).

In the southern part of the basin, the scarcity of chronostratigraphic constraints makes it more challenging to correlate sections, but the top ~190 m intervals of sections S01 and S02 feature generally coarser strata that may be correlative to the unit defined as Em₄ in the north. These strata consist of medium- to thick-bedded sandstone (FA 7) and interbedded mudrock (FA 4; Fig. 6G), forming an overall upward-coarsening sequence. The sandstone layers in FA 7 are also tabular, laterally continuous over hundreds of meters, and characterized by St (Fig. 6F), Sp, and Sr lithofacies. In certain horizons, the sandstones show heavy bioturbation (Sb). Carbonate and gypsum beds are less common in unit Em₄, with only one thinly bedded carbonate layer documented in section S01.

Paleoflow directions were reconstructed using cross-bedding measurements at different locations, with at least five measurements averaged per site. The results indicate a southward paleoflow in the northern part of the basin and a northward paleoflow in the southern part of the basin.

4.1.4.2. Interpretation. The red strata containing FA 4 and FA 5 in the lower 306 m are

interpreted as having been deposited in a deltaic plain, influenced by both river discharge and lake-level fluctuations. The mudrock and carbonate in FA 4 were likely deposited during periods of reduced river clastic input, coinciding with lake-level highstands. In contrast, the well-organized sandstone and conglomerate in FA 5 likely represent cut-and-fill deposition in unidirectional river channels on the deltaic plain, while the less common poorly organized conglomerate and sandstone beds may reflect deposition during flood events.

In the upper 472 m of unit Em₄, the tabular gray layers with lateral continuity, comprising structureless sandstone and laminated mudrock in FA 6, are interpreted as gravity-flow deposits in a delta-front to prodelta setting of a relatively deep lacustrine environment. These deposits resemble hybrid event beds, which are deposited by flows exhibiting “mixed behavior,” resulting in complex successions of sandstone and mudrock (Baas et al., 2011; Stevenson et al., 2020). The Sm at the base represents the H1 of the hybrid event beds; the banded sandstone represents H2, which formed through bed-form development under transitional flow conditions; the mud-rich sandstone with mud clasts is typical of hybrid event beds and represents H3, with

TABLE 2. SUMMARY OF TUFF AND CARBONATE AGES FROM THE XIANQIAN BASIN

Tuff sample no.	Lithology	Unit	Location	Global positioning system location	Age (Ma)	Data source
23XQ316	Tuff	Em ₃	Section CW01	33.0893°N, 83.0659°E	45.23 ± 0.40 (<i>n</i> = 15, MSWD = 0.87)	This study
23NR93	Carbonate	Em ₃	Section NE01	33.0347°N, 83.5373°E	47.8 ± 5.1 (<i>n</i> = 35, MSWD = 7.4)	This study
23NR56	Tuff	Em ₃	Southern Xianqian Basin	32.9347°N, 83.0775°E	46.32 ± 0.54 (<i>n</i> = 15, MSWD = 2.4)	A. Ma (2025, personal commun.)
23GZ60	Tuff	88 m,† Em ₁	Section N01	33.1476°N, 83.1112°E	49.07 ± 0.37 (<i>n</i> = 15, MSWD = 1.1)	A. Ma (2025, personal commun.)
23JC139	Tuff	930 m,† Em ₄	Section N01	33.1393°N, 83.1068°E	44.53 ± 0.70 (<i>n</i> = 15, MSWD = 3.3)	A. Ma (2025, personal commun.)
23GZ78	Tuff	1016 m,† Em ₄	Section N01	33.1389°N, 83.1048°E	43.46 ± 0.80 (<i>n</i> = 13, MSWD = 2.1)	A. Ma (2025, personal commun.)
21AW40*	Tuff	706 m,† Em ₄	Section N02	33.1503°N, 83.0819°E	44.67 ± 0.18 (<i>n</i> = 4, MSWD = 0.052)	A. Ma (2025, personal commun.)
21AW45	Tuff	953 m,† Em ₄	Section N02	33.1502°N, 83.0816°E	44.38 ± 0.25 (<i>n</i> = 22, MSWD = 3.5)	A. Ma (2025, personal commun.)
21AW48	Tuff	994 m,† Em ₄	Section N02	33.1504°N, 83.0806°E	43.98 ± 0.20 (<i>n</i> = 22, MSWD = 2)	A. Ma (2025, personal commun.)
21AW51*	Tuff	1037 m,† Em ₄	Section N02	33.1502°N, 83.0783°E	43.67 ± 0.15 (<i>n</i> = 4, MSWD = 0.38)	A. Ma (2025, personal commun.)
23XQ215	Tuff	1178 m,† Em ₄	Section N02	33.1509°N, 83.0779°E	43.39 ± 0.51 (<i>n</i> = 23, MSWD = 2.1)	A. Ma (2025, personal commun.)
23XQ247	Tuff	1395 m,† Em ₄	Section N02	33.1387°N, 83.1065°E	43.12 ± 0.55 (<i>n</i> = 8, MSWD = 0.63)	A. Ma (2025, personal commun.)
23XQ266	Tuff	1652 m,† Em ₄	Section N03	33.1589°N, 83.0644°E	43.30 ± 1.20 (<i>n</i> = 13, MSWD = 8.2)	A. Ma (2025, personal commun.)

Note: MSWD—mean square of weighted deviates.

*Zircons from these two tuff samples were dated with the chemical abrasion–isotope dilution–thermal ionization mass spectrometry (CA-ID-TIMS) method, and those from other tuff samples were dated with secondary ion mass spectrometry (SIMS).

†Sample horizon at the composite section on Figure 3.

subunit H3b containing more mud than H3a; the thin, graded sandstone and mudrock with common parallel and ripple lamination represent H4, which is related to low-density flows in the turbulent wake; and the mudrock cap represents H5, which is formed by suspension fall-out (Haughton et al., 2009; Hussain et al., 2020; Dodd et al., 2022). The common association of these beds with lithofacies Vt and especially Vb suggests that the influx of coarse-grained deposits may have been associated with syndepositional volcanism, with the presence of volcanoclastic breccia indicating nearby volcanic eruptions.

FA 7 in the southern Xianqian Basin (sections S01 and S02), characterized by medium- to thick-bedded layers of St, Sp, and Sh, was likely deposited through tractional flows in distributary channels in a delta front. The upward-coarsening packages probably represent progradational deltaic sequences in shallow lakes (e.g., Bohacs et al., 2003). The red siltstones (F) and bioturbated sandstone (Sb) within FA 4 represent fine-grained overbank deposits on delta plains. The carbonate bed within FA 4 was likely deposited during a lake-level highstand, when clastic detrital input was reduced.

4.1.5. Unit Em₅: Upper Conglomerates

4.1.5.1. Description. Unit Em₅ (upper conglomerates) is >188 m in thickness and only present in section N03 in the northern Xianqian Basin. It includes FA 8, which is dominated by boulder to pebble conglomerates, with local occurrences of sandstone, mudrock, and tuff beds. The conglomerate is clast-supported and generally massive (Gcm; Fig. 6H). Clasts reach up to 40 cm in diameter, are poorly sorted, and are moderately rounded at the base of the unit. Moving up section, the conglomerate becomes more organized, exhibiting horizontal stratification (Gch). Clast size decreases upward to pebbles that are better rounded and better sorted than boulders and pebbles below. The conglomerates are interbedded with 10–50-cm-thick beds of sandstone (Sm), tuff (Vt), and red mudrock (F).

4.1.5.2. Interpretation. FA 8 in unit Em₅ is interpreted as having been deposited in a gravity flow–dominated alluvial-fan environment. Compared to FA 1 in unit Em₁, the massive conglomerate occurs more frequently in FA 8, and the clasts are larger in size overall. The Gcm and Sm lithofacies indicate deposition by high-density debris flows, while Gch may represent sieve deposits on an alluvial fan.

4.2. Chronostratigraphic Controls

Zircon U–Pb dating in this study suggests the volcanic rocks bounding the Xianqian Basin are late Early Cretaceous in age. In sections SW01 and SE01, the felsic volcanic rocks unconformably underlying the Maerguo Formation yielded zircons with weighted mean U–Pb ages of 108.8 ± 2.6 Ma (*n* = 19, MSWD = 6.2, sample 23NR50) and 110.0 ± 1.7 Ma (*n* = 18, MSWD = 1.3, sample 23GZ41), respectively (Figs. S1A–S1B). In section NE01, the felsic volcanic rock in the hanging wall yielded a zircon U–Pb age of 108.1 ± 1.8 Ma (*n* = 20, MSWD = 4.2, sample 23NR83; Fig. S1C).

The age of the Maerguo Formation was determined through zircon U–Pb dating of tuff samples and carbonate U–Pb dating (Table 2), from both this study and from A. Ma (2025, personal commun.). In this study, the tuff sample 23XQ316 from unit Em₃, located at the top of section CW01, yielded a weighted mean zircon U–Pb age of 45.2 ± 0.4 Ma (*n* = 15, MSWD = 0.87; Fig. S1D). In section NE01, the carbonate sample 23NR93 from unit Em₃ yielded a lower-intercept carbonate U–Pb age at 47.8 ± 5.1 Ma (*n* = 35, MSWD = 7.4) on the Tera–Wasserburg

plot (Fig. S1E). For sedimentary rocks in other sections that were not dated with tuff zircon or carbonate U–Pb methods, stratigraphic correlation was based on geologic mapping, lithology, and sedimentary facies.

4.3. Sediment Provenance Analysis

4.3.1. Modal Sandstone Petrology

Twenty-six sandstone samples from unit Em₂ in both the northern and southern Xianqian Basin, eight sandstone samples from units Em₃–Em₄ in the southern Xianqian Basin, and one sandstone sample in the lowermost unit Em₃ (23GZ64) in the northern Xianqian Basin are rich in quartz grains and plot mostly in the feldspatho-litho-quartzose field of the quartz-feldspar-lithics (QFL) diagram (Fig. 12), following the classification scheme of Garzanti (2019). The lithic fragments are dominated by felsic volcanic rocks and limestone, with subordinate and varying proportions of slate, phyllite, and schist (Fig. 12).

In the northern Xianqian Basin (sections N01–N04), 38 samples from unit Em₄ and uppermost unit Em₃ record a major change in provenance compared to unit Em₂ (Fig. 12). From unit Em₂ to unit Em₃, there is an increase in feldspar and lithic fragments at the expense of quartz, and an increase in metamorphic fragments at the expense of sedimentary fragments. Overall, higher in the stratigraphic level, there is a greater abundance of schist fragments and plutonic fragments. Unit Em₄ in section NW01 and one sample (23NR41) from unit Em₃ in section CW01, both located in the northwest Xianqian Basin, show similar quartz content but a higher concentration of limestone fragments compared to sections N01–N04. This indicates a provenance variation along strike along the northern margin of the Xianqian Basin.

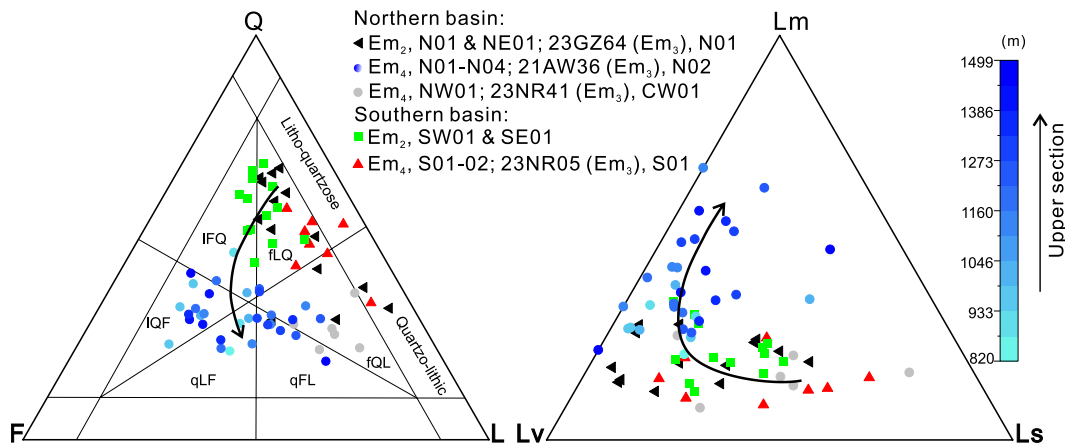


Figure 12. Ternary diagram showing the sandstone petrography in QFL plot with sandstone classification scheme after Garzanti (2019). Q—quartz; F—feldspar; L—lithic fragments (Lv—volcanic; Ls—sedimentary; Lm—metamorphic); fLQ—feldspatho-litho-quartzose; fQL—feldspatho-quartzo-lithic; qFL—quartzo-feldspatho-lithic; IFQ—litho-feldspatho-quartzose; IQF—litho-quartzo-feldspathic; qLF—quartzo-lithofeldspathic; qFL—quartzo-feldspatho-lithic.

4.3.2. Conglomerate Clast Counts

Clasts in conglomerate were counted at 10 sites throughout the entire stratigraphic thickness of the Maerguo Formation (Figs. 7–10). In unit Em₁ of the Maerguo Formation, clasts are predominantly composed of limestone (35%–62%, averaging 51%), felsic volcanic rock (20%–41%, averaging 28%), sandstone (11%–16%, averaging 13%), and low-grade metamorphic rock (5%–12%, averaging 8%). Metamorphic clasts are largely absent in unit Em₂ along the southern margin of the basin and in unit Em₄ in section NW01, where the clasts are dominated by felsic volcanic rock (22%–83%, averaging 52%), sandstone (11%–41%, averaging 29%), and limestone (0%–34%, averaging 15%), with insignificant amounts of quartz and mudrock. In contrast, single-site clast counts of unit Em₄ in section N02 and unit Em₅ in section N03 revealed significant proportions of low-grade metamorphic clasts (41% and 18%, respectively), granitoid (12% and 38%, respectively), mudrock (11% at both sites), sandstone (13% and 10%, respectively), and limestone (6% and 23%, respectively), with quartz (5%) and felsic volcanic rock (12%) present only in unit Em₄.

4.3.3. Detrital Heavy Minerals

The heavy mineral content in units Em₂ and Em₃ in the entire Xianqian Basin and unit Em₄ in the southern Xianqian Basin ranges between 0.3% and 2.1% with an average of 0.9%. The transparent heavy mineral suite accounts for 4%–68% (with an average of 28%) in volume and consists of zircon, tourmaline, apatite, garnet, rutile, titanite, epidote, and rare grains of amphibole and clinopyroxene. In the biplot of heavy mineral data, these samples plotted mainly in the first and fourth quartiles, indicating dominance of zircon, rutile, tourmaline, and titanite, among others (Fig. 13). The ZTR index varies between 27 and 78 with an average of 58.

Unit Em₄ in the northern Xianqian Basin yielded notably different heavy mineral characteristics, with a higher proportion of transparent heavy minerals compared to strata below the unit and elsewhere in the Xianqian Basin. The heavy mineral content varies between 0.3% and 4.5% (average of 2.5%), and the transparent heavy mineral suite accounts for 33%–90% (average of 64%). Compared to that of unit Em₂ underneath, the transparent heavy mineral suite is characterized by an increasing content of amphibole at the expense of apatite, zircon, tourmaline, and rutile (Fig. 13). Within unit Em₄ in the northern Xianqian Basin, the ZTR index declines upward from 38 to insignificant values, while amphibole comprises up to 94% in the uppermost unit Em₄.

4.3.4. Detrital Zircon U-Pb Age Spectra

Figure 14 presents kernel density estimation (KDE), probability density plot (PDP), and histogram (bin width 10 Ma) representations of the detrital zircon U-Pb ages from sandstones in the Xianqian Basin and potential source terranes. KDE is preferred here for detrital provenance interpretation due to its solid theoretical underpinnings and its ability to avoid the double-smoothing effect and artificial peaks found in PDPs, resulting in more accurate probability density estimates (Vermeech, 2012). Three sandstone samples in unit Em₂, including one from section NE01 (23NR84) in the northern Xianqian Basin and two (23GZ46 in section SE01, 23NR54 in section SW01) in the southern basin, and one sandstone sample (23NR18) in unit Em₄ from the southern Xianqian Basin share similar zircon U-Pb age spectra, including an age peak ca. 110 Ma (Fig. 14). Older peaks at 150 Ma, 220 Ma, 425 Ma, 860 Ma, 1870 Ma, and 2500 Ma are present in 23NR84, 23GZ46, and 23NR18 but are insignificant in 23NR54 in the southwestern Xianqian Basin (Fig. 14). Samples 23NR84, 23GZ46, and 23NR18 con-

tained a few scattered ages between 60 Ma and 100 Ma.

In the northern Xianqian Basin, detrital zircons within unit Em₂ sandstone (23GZ61 in section N01) are characterized by multiple U-Pb age peaks at 115 Ma, 160 Ma, 220 Ma, 270 Ma, 1000 Ma, 1850 Ma, and 2500 Ma (A. Ma, 2025, personal commun.), which are very similar to those of 23NR84 in the northeast Xianqian Basin along strike (Fig. 14). In contrast, samples from unit Em₄ (21AW49 in section N02, 23DG84.5 in section N04, and 23XQ258 in section N03) in the northern Xianqian Basin are characterized by a single dominant age peak at 160–150 Ma (A. Ma, 2025, personal commun.; Fig. 14). A granitoid clast (sample 23XQ271) in unit Em₅ yielded a weighted mean zircon U-Pb age of 161.5 ± 2.7 Ma ($n = 19$, MSWD = 2.8; see Fig. S1F).

To investigate potential sources for the basin, we analyzed several samples outside of the basin. A sandstone sample (23NR60) from Jurassic strata to the north of the Xianqian Basin yielded detrital zircons with U-Pb ages peaks at 200 Ma, 270 Ma, 520 Ma, and 1130 Ma, multiple small peaks between 550 Ma and 1040 Ma, and other insignificant small peaks (Fig. 14). The maximum depositional age of this Jurassic sandstone is constrained by the three youngest detrital zircons dated: 179 ± 4 Ma, 181 ± 3 Ma, 188 ± 4 Ma. One sample (23TG01) from the Cretaceous Abushan Formation continental red bed and one sample (23TG14) previously mapped as the Upper Triassic Tinggongcuo Formation (Yue et al., 2006) yielded similar detrital zircon U-Pb age spectra, both of which displayed scattered multiple age peaks at 300–200 Ma and 1000–500 Ma, along with subordinate peaks at 1840 Ma and 2500 Ma (Fig. 14). The youngest age peak in both stratigraphic units is Late Jurassic (23TG01 at ca. 160 Ma and 23TG14 at ca. 145 Ma). The existence of abundant zircons

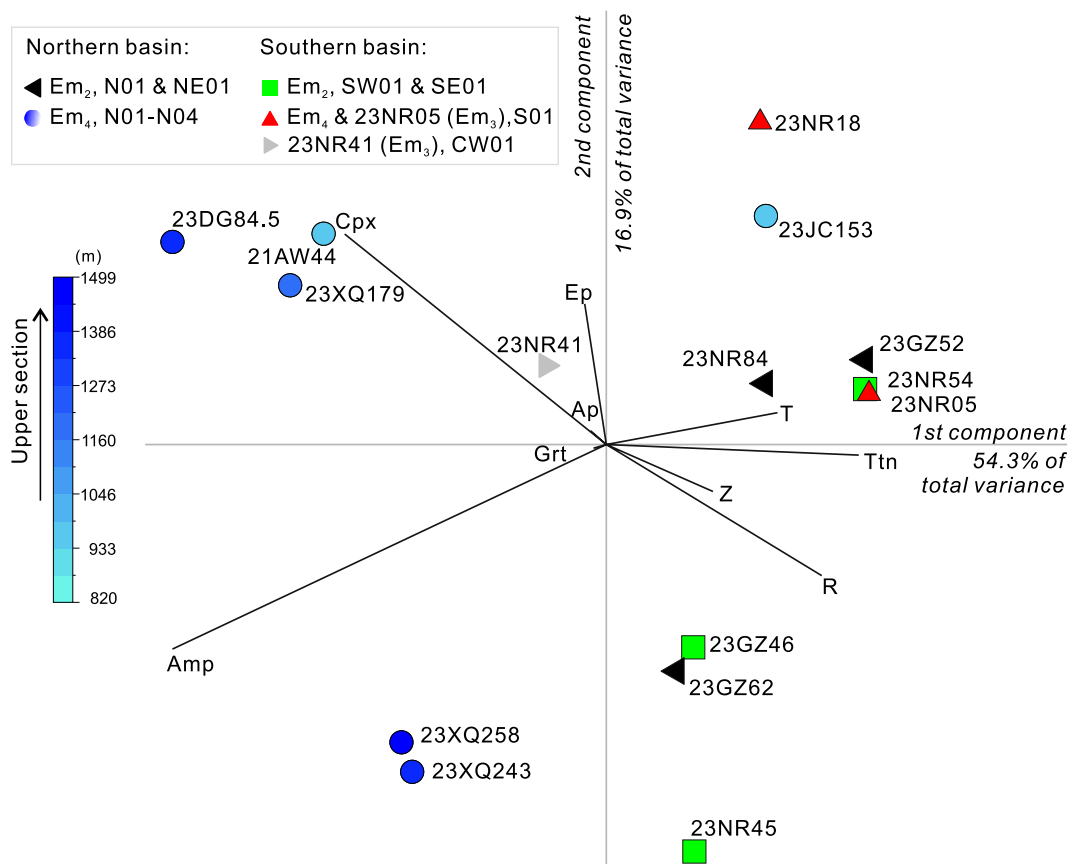


Figure 13. Heavy mineral compositional biplot. The length of each ray is proportional to the variance of the corresponding element in the dataset; if the angle between two rays is close to 0°, 90°, or 180°, then the corresponding variables are directly correlated, uncorrelated, or inversely correlated, respectively (Gabriel, 1971). Ap—apatite; Cpx—clinopyroxene; Grt—garnet; Ttn—titanite; Ep—epidote; Amp—amphibole; Z—zircon; T—tourmaline; R—rutile.

with ages near the Jurassic-Cretaceous boundary suggests a Cretaceous or younger depositional age for the so-called Tinggongcuo Formation. In addition, 350 Ma, 390 Ma, and 435 Ma peaks in 23TG01 are insignificant in 23TG14, which instead exhibits a prominent 450 Ma peak (Fig. 14).

4.4. Tectonic Subsidence of the Xianqian Basin

The reconstructed tectonic subsidence curve (Fig. 3) suggests the Xianqian Basin experienced ~710 m of tectonic subsidence from ca. 50 Ma to 43 Ma, with a relatively low tectonic subsidence rate at ~0.1 mm/yr. Then, tectonic subsidence accelerated to ~0.7 mm/yr, and the basin subsided an additional 260 m by ca. 43.3 Ma (Fig. 3). Finally, tectonic subsidence decelerated back to <0.2 mm/yr with a total of 110 m of tectonic subsidence by ca. 42.6 Ma (Fig. 3).

5. DISCUSSION

5.1. Sediment Provenance Interpretation

The southern Xianqian Basin records no significant provenance changes throughout the

depositional history of the Maerguo Formation, whereas the northern Xianqian Basin shows a provenance shift between units Em₁–Em₂ and units Em₄–Em₅. Units Em₁, Em₂, and Em₄ in the southern Xianqian Basin exhibit a recycled orogen provenance (Dickinson, 1985; Garzanti et al., 2007), as the sandstones from these units are dominated by quartz and lithic fragments, with the latter primarily consisting of felsic volcanic and sedimentary fragments (Fig. 12). Conglomerate compositions generally support this interpretation, with clasts predominantly composed of sandstone, volcanic rocks, limestone, and minor amounts of quartz and mudrock. The dominance of zircon, tourmaline, apatite, garnet, and rutile in the heavy mineral assemblage, with an average ZTR index of 58, also supports the recycled provenance interpretation. In certain stratigraphic levels of sections SE01 and SW01, volcanic clasts dominate, likely recycled from Cretaceous volcanic rocks exposed farther to the south. The youngest detrital zircon U–Pb ages, peaking at ca. 110 Ma, further confirm a contribution from Cretaceous volcanic rocks. The presence of multiple older age peaks at ca. 160 Ma, 220 Ma, 270 Ma, 1000 Ma, 1850 Ma, and 2500 Ma suggests recycling of materials from pre-Cretaceous sedimentary rocks that

were exposed around the basin during deposition (Li et al., 2017; Ma et al., 2017).

Compared to units Em₂–Em₃, units Em₄–Em₅ in the northern Xianqian Basin exhibit a significant change in detrital composition, characterized by increased feldspar content and decreased quartz content in sandstones (Fig. 12). This shift is typical of rocks derived from a transitional to dissected magmatic arc (Dickinson, 1985). The presence of granitoid fragments, a distinct detrital zircon U–Pb age peak at ca. 150 Ma in sandstone, ca. 162 Ma granitoid clasts in conglomerate, and the significant amount of hornblende in the heavy mineral assemblage further support this interpretation. However, the abundance of low-grade metamorphic rock, mudrock, sandstone, volcanic rock, and limestone fragments in both sandstone and conglomerate points to a recycled orogen source. U–Pb age signals of other contributions may have been masked by the prominence of granitoid rocks, which are rich sources of sand-sized zircon (Garzanti and Andò, 2019). Based on paleocurrent measurements, we suggest the major source of sediment was the Late Jurassic granitoids located to the north of the basin, which is further supported by the abundance of amphibole and clinopyroxene in heavy minerals against the

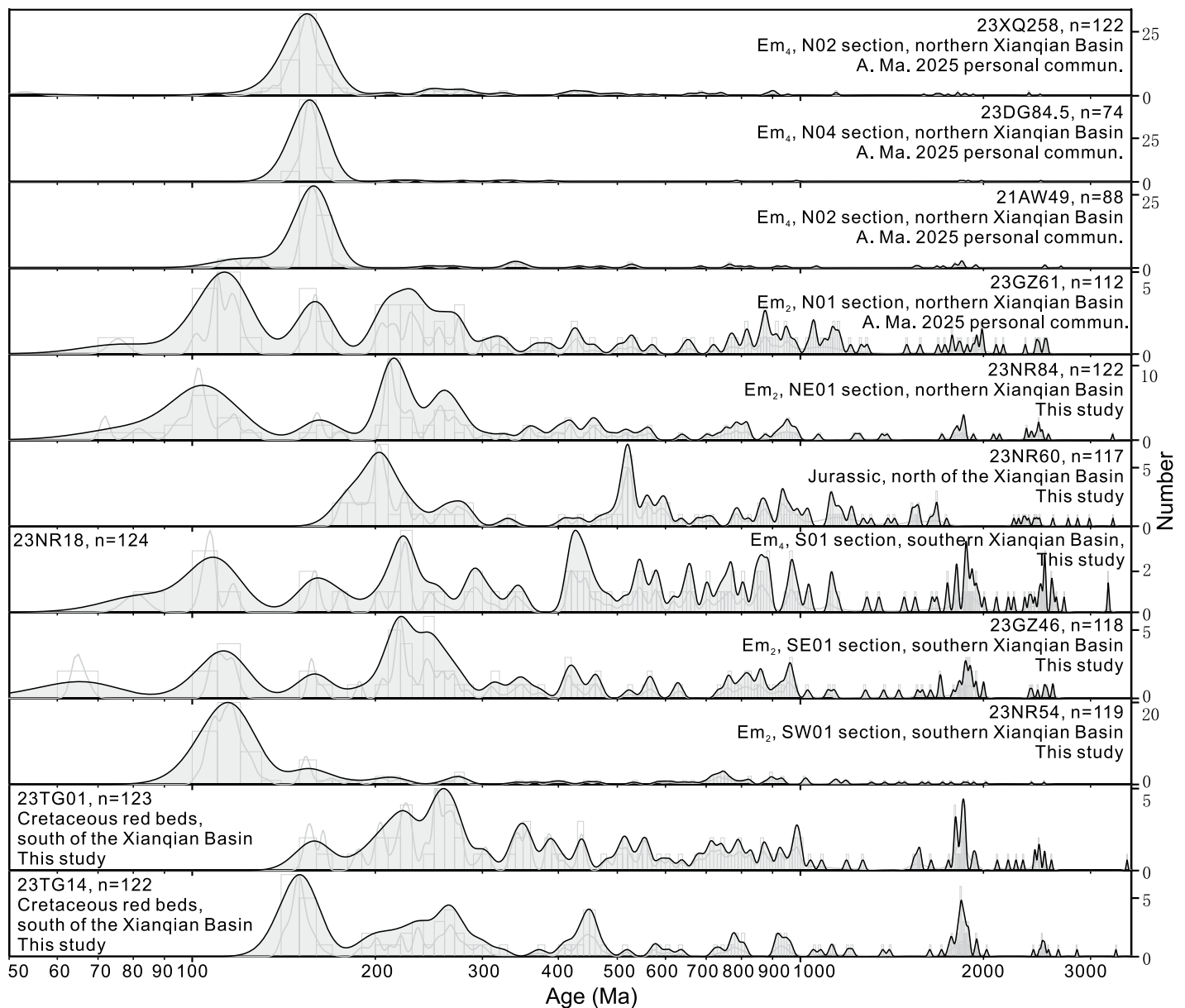


Figure 14. Kernel density estimation (KDE; black curve, filled, and bandwidth 50 Ma), probability density plot (PDP; gray curve), and histograms (bin width 10 Ma) of the detrital zircon U-Pb ages from the sandstones of potential provenance and from the sandstones in the Xianqian Basin.

percentage of 166–143 Ma detrital zircon ages within the Maerguo Formation (Fig. S3). This provenance change, documented in the northern Xianqian Basin at ca. 45 Ma, implies more rapid unroofing of the source area to the north of the basin, or the exhumation of granitoids that were previously unexposed.

5.2. Basin Architecture and Paleogeography

The stratigraphy, sedimentology, and provenance analyses demonstrate that the oval-shaped Xianqian Basin is dominated by

coarse-grained facies along the margins and fine-grained lacustrine facies in the center. Figure 15 depicts a tentative lithostratigraphic correlation based on our measured sections. We suggest that the Xianqian Basin was filled by three distinct depositional systems: an initial lower alluvial-fluvial system with bidirectional paleoflows, coinciding with north-south shortening that induced the development of growth strata in the northern basin; a medial lacustrine system into which marginal deltas systems prograded, again characterized by bidirectional provenance; and an upper alluvial system in the

northern basin, primarily sourced from granitoid and metamorphic rocks in the hanging wall of the Xianqian thrust.

The basin fill exhibits an overall upward fining followed by a coarsening grain-size trend, resembling the classic “lacustrine sandwich” pattern, which is often considered a hallmark of extensional basins (Schlische, 1992). However, the presence of the syndepositional north-dipping Xianqian thrust and growth strata in the north suggests that the Xianqian Basin developed in a contractional setting. This contraction did not significantly affect the southern Xianqian Basin,

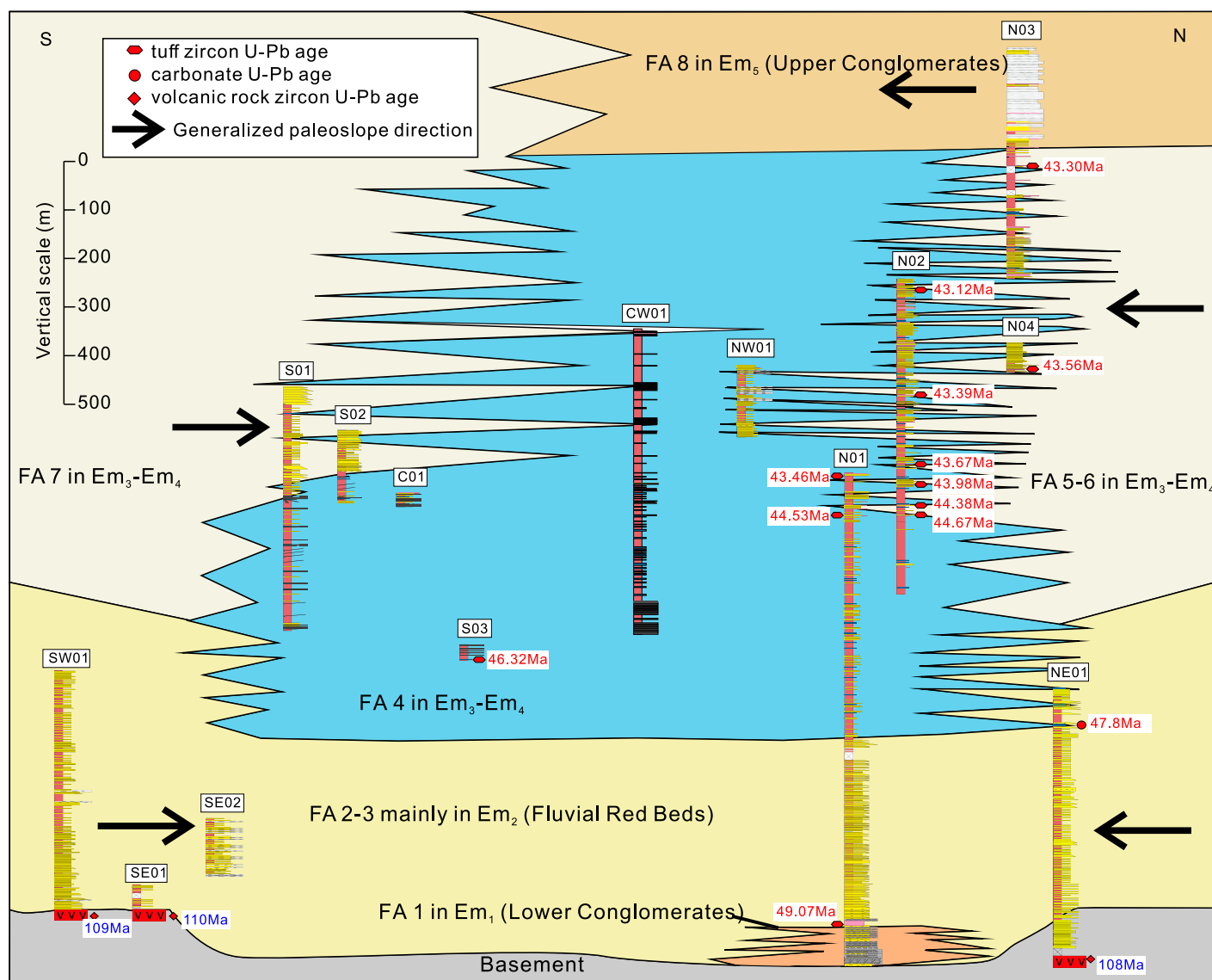


Figure 15. Tentative correlation diagram for measured sections in the Maerguo Formation. The locations of sections can be found in Figure 2A. Correlations were established based on marker beds and tuff geochronology. Generalized paleoslope directions are plotted based on paleocurrent measurements. FA—facies association.

where the Maerguo Formation rests unconformably upon Cretaceous volcanic and Jurassic sedimentary rocks and dips gently to the north.

One interpretation is that the Xianqian Basin was a foreland basin, and the subsidence was controlled by the southward propagation of a thrust belt. This interpretation is consistent with the acceleration of subsidence over time, followed by basin inversion, which could be interpreted to be the transition from forebulge to foredeep and finally to wedge top, related to southward propagation of a thrust load (DeCelles and Giles, 1996). At the same time, the Xianqian Basin is notably unlike other thrust-bounded basins in Tibet, which are elongate and narrow

(<15 km in width) in map view, and dominated by kilometers-thick sections of upward-coarsening conglomeratic sequences associated with basin-scale growth strata and overturned synclines (Horton et al., 2002; Kapp et al., 2007; DeCelles et al., 2007; Li et al., 2018). The widespread deposition of the basal fluvial unit around the beginning of deposition in the Xianqian Basin (over >40 km of the basin from north to south, before restoration of any Eocene and younger shortening) suggests a broad area of subsidence that potentially requires another tectonic driver. One possibility is dynamic subsidence due to the localized removal of dense lower lithosphere (e.g., DeCelles et al., 2015;

He and Kapp, 2023; McMillan and Schoenbohm, 2023), which would also be consistent with the observed subsidence curve (Fig. 3), stratigraphic patterns, rapid postburial exhumation, and increase in proximal volcanism around the time of maximum basin subsidence (J. He, 2025, personal commun.). We consider this latter possibility in more detail in a companion paper addressing the structural geology of the basin (J. He, 2025, personal commun.).

Based on stratigraphy, sedimentology, provenance, and section correlation, the paleogeographic evolution of the Xianqian Basin can be divided into four stages (Fig. 16). During the first stage between 49.6 Ma and 47.0 Ma,

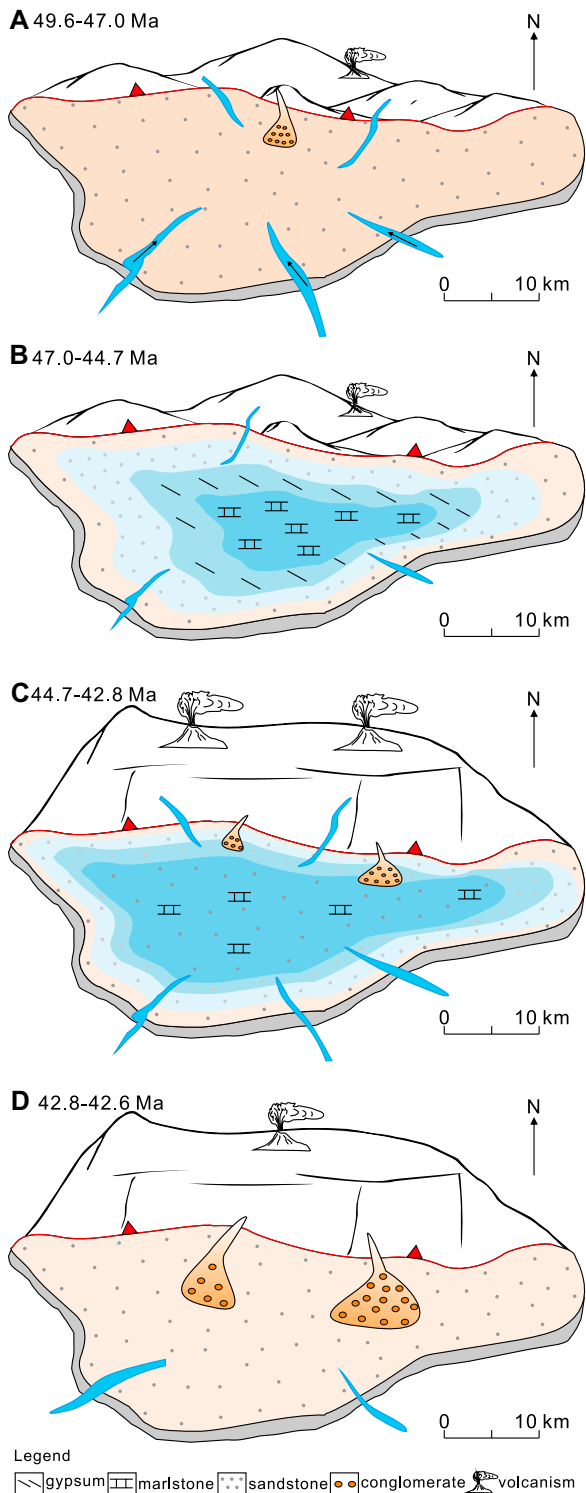


Figure 16. Paleogeography and paleoenvironment of the Xianqian Basin at four periods from ca. 50 Ma to 43 Ma. See text for discussion.

enance shift was accompanied by a threefold increase in sediment accumulation rate from 18.0 cm/k.y. to 71.9 cm/k.y. and enhanced tectonic subsidence rate (from ~ 0.1 mm/yr to ~ 0.7 mm/yr; Fig. 3). Magmatic activity also intensified during the deposition of unit Em₄. The magmatism is interpreted as proximal, as the tuff breccias include abundant granitoid and metamorphic clasts that were likely sourced from local areas to the north of the Xianqian Basin. However, magmatic rocks of 50–43 Ma age have not been reported around the basin, possibly due to complete erosion or insufficient geologic exploration. The southern margin of the Xianqian Basin recorded rejuvenated sand sediment flux, but no significant change in provenance (Fig. 16C), as recorded in unit Em₄. During the fourth stage, from 42.8 Ma to 42.6 Ma, the basin was filled quickly, with coarse-grained detritus deposited in an alluvial-fan environment in the northern basin (Fig. 16D). The paleoenvironment in other parts of the basin is unknown because the uppermost unit of the formation is not exposed elsewhere.

5.3. Implications for Paleoenvironments and Tibet Uplift

Deposition of abundant lacustrine gypsum and limestone in unit Em₃, along with inward paleoflow directions, suggests that the Xianqian Basin was an internally drained saline lake basin. The disappearance of gypsum, coupled with the presence of punctuated dark-green to gray mudrock and sandstone with leaf and insect fossils in unit Em₄ (see Results section; Figs. 11C and 11D), likely indicates a transition to a less arid environment.

Compared to the short-lived Xianqian Basin during 50–43 Ma, the Bangong-Nujiang suture basins were longer-lived (Fig. 17). In the Gerze Basin, the Kangtuo Formation was deposited during ca. 55–40 Ma and dominated by red conglomerates and sandstones, likely also indicating a relatively arid environment (Song et al., 2023). In the Nima-Lunpola-Bangoin basins to the east, coarse deposits of the lower Niubao Formation were deposited earlier than 50 Ma and as young as 40 Ma (Xiong et al., 2022), and the mudrocks in the Bangoin Basin contain abundant biota suggesting subtropical wet environments at ca. 47 Ma (Su et al., 2020). Clumped isotopes of paleosol carbonate in Lunpola (Xiong et al., 2022) and leaf morphotypes in Bangoin (Su et al., 2020) both suggest <2 km paleoelevation.

The Gerze Basin to the south of the Xianqian Basin was suggested to be near sea level around the late Eocene (ca. 36 Ma) based on the presence of marine foraminifera in fine-grained lacustrine strata of the Suonahu Formation (Wei

units Em₁ and Em₂ were deposited. An alluvial-fan environment was developed locally in the northern Xianqian Basin, while fluvial systems dominated along both the northern and southern margins, with bidirectional paleodrainage feeding the basin (Fig. 16A). During the second stage from 47.0 Ma to 44.7 Ma, a lake expanded,

resulting in the deposition of limestone, gypsum, and mudrock (Fig. 16B). During the third stage between 44.7 Ma and 42.8 Ma, the lake deepened, and sediments sourced from the Jurassic granitoids and metamorphic rocks to the north of the Xianqian Basin were deposited in deltaic settings along the northern margin. This prov-

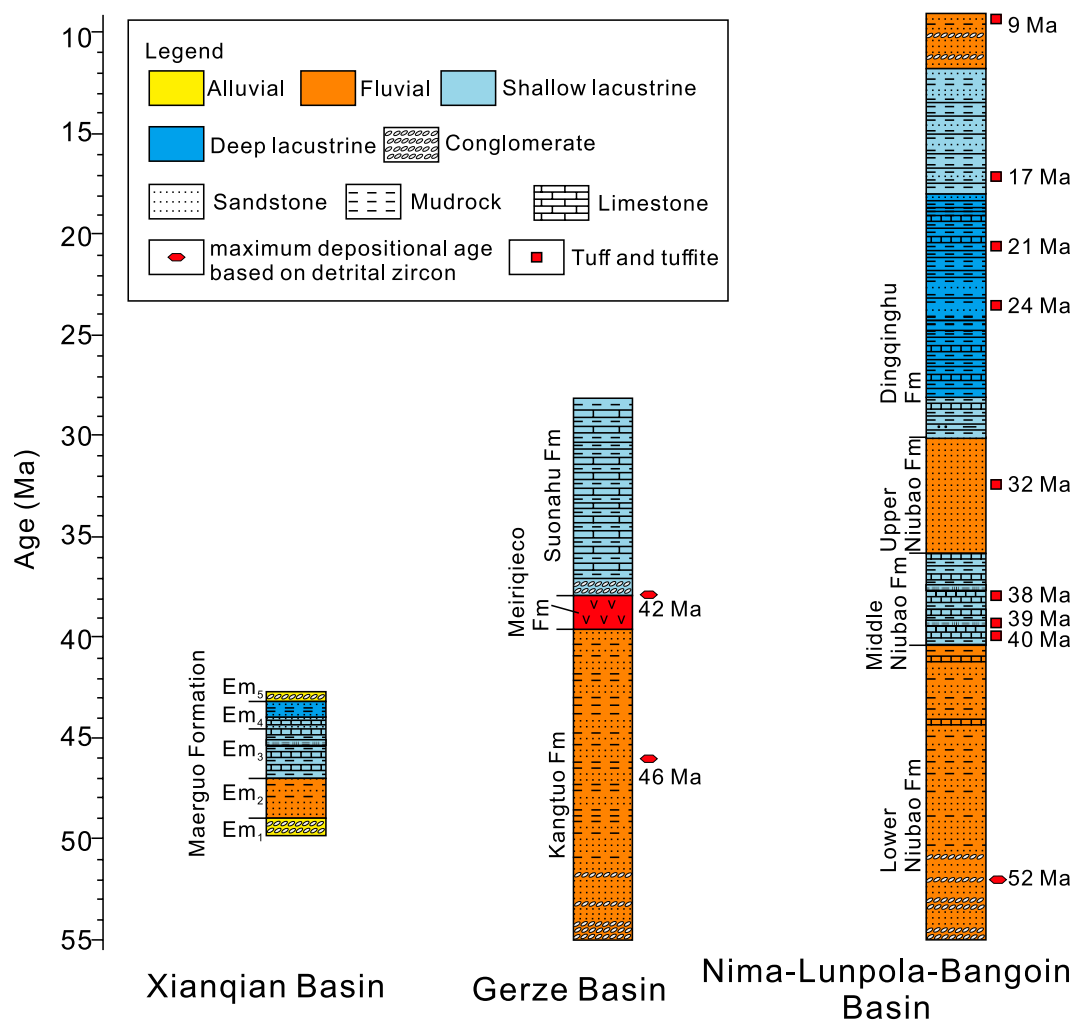


Figure 17. Stratigraphy of the Xianqian Basin in comparison with Gerze (Wei et al., 2016; Song et al., 2023) and Nima-Lunpola Basins (Han et al., 2019; Fang et al., 2020; Xiong et al., 2022).

et al., 2016). The relatively positive $\delta^{18}\text{O}$ values (-6‰ to -4‰ relative to Vienna Pee Dee belemnite [VPDB]) of the lacustrine carbonate were interpreted to indicate low paleoelevations close to sea level (Wei et al., 2016) or high paleoelevations due to evaporation (Ingalls et al., 2020). However, recent clumped and oxygen isotopic datasets indicate the existence of shallow burial dolomitization and late-stage telogenetic carbonate precipitation after exhumation to near Earth's surface, which complicates paleoelevation interpretation (Li et al., 2024a). From a geologic perspective, the Gerze Basin should have been lower in paleoelevation than the sediment source area to its north during ca. 55–30 Ma, when the Kangtuo and Suonahu Formations were deposited (Song et al., 2023). In addition, as the Xianqian Basin was deposited on rocks in the hanging wall of the Shiquanhe-Gerze-Amdo thrust system, which was active as early as Early Cretaceous time (Kapp et al., 2005; Li et al., 2022a), its paleoelevation may have been higher than the cotem-

poraneous Gerze Basin, which was deposited in the footwall.

Based on regional basin evolution, we present schematic north-south cross sections to show the four-stage topographic history of central Tibet (Fig. 18). During the first stage, from 49.6 Ma to 47.0 Ma, the Xianqian Basin started to subside (~ 0.1 mm/yr), and the north-dipping Xianqian thrust in the north activated, possibly associated with growth of a dense lithospheric instability beneath the basin (Fig. 18A) in conjunction with north-south-directed crustal shortening. Scattered magmatism occurred in the Qiangtang interior (Fig. 1B). The Gerze Basin along the Bangong-Nujiang suture zone, developing in the footwall of the Shiquanhe-Gerze-Amdo thrust, was lower in elevation than the Xianqian Basin in the hanging wall (Fig. 18A). Farther south, the southern Lhasa terrane had already risen to >4 km while the Indian lithosphere was subducting northward beneath it (Ding et al., 2014; Ingalls et al., 2018; Ibarra et al., 2023). Then from 47.0 Ma to

44.7 Ma, the Xianqian Basin deepened slightly to form a shallow lake environment. Scattered tuff beds indicate continued but volumetrically minor volcanic activity in the Qiangtang interior (Fig. 18B). During these two stages, the basin subsided at a relatively low rate (Fig. 3), which would be consistent with the slow subsidence expected at the early stages of basin formation associated with lithospheric drips, as also observed in the Arizario Basin in the Puna Plateau (DeCelles et al., 2015).

From 44.7 Ma to 42.8 Ma, the Xianqian Basin further deepened and was filled with delta-front to prodelta gravity-flow deposits. The basin experienced an acceleration in tectonic subsidence rate from ~ 0.1 mm/yr to ~ 0.7 mm/yr at 43.7 Ma, before decelerating to <0.2 mm/yr at ca. 43.3 Ma. The basin subsidence acceleration would be consistent with the acceleration in strain rate as the dense lithosphere root began to sink (He and Kapp, 2023). The increasingly rapid growth and descent of the lithosphere root would cause

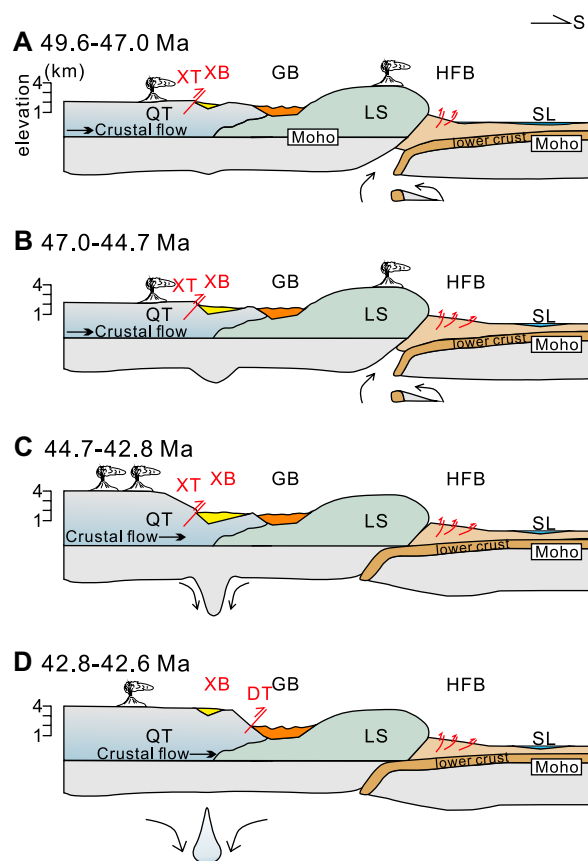


Figure 18. Schematic north-south cross sections show basin evolution during four different stages in the Eocene, which is proposed to have been controlled at least in part by southward propagation of thrust belts in southern Qiangtang and possibly dripping of dense lower lithosphere, as discussed in the text. The sea retreat history and evolution of the Himalaya foreland basin system are after Zhang et al. (2025). LS—Lhasa terrane; QT—Qiangtang terrane; XT—Xianqian thrust; DT—Dongco thrust; XB—Xianqian Basin; GB—Gerze Basin; HFB—Himalaya foreland basin; SL—sea level.

upwelling of asthenosphere and cause partial melting (Wang and Currie, 2015). Volcanism may have occurred in areas proximal to the Xianqian Basin, and this was preserved in the basin as tuff beds. The inferred magmatic heating and addition, coupled with isostatic uplift from crustal thickening due to thrusting (e.g., the Xianqian thrust) and possibly crustal flow, may have elevated the Qiangtang plateau north of the Xianqian Basin from ~2 km to >4 km (Fig. 18C; A. Ma, 2025, personal commun.).

Tectonic subsidence remained low (<0.2 mm/yr) from 43.3 Ma to 42.6 Ma, but during 42.8–42.6 Ma, the Xianqian Basin was rapidly infilled by alluvial-fan deposits. If the basin formed as a result of lithospheric drip, detachment of the drip would have led to dynamic rebound and isostatic uplift, which could have inverted the basin and resulted in proximal coarse deposition in unit Em₅. The Dongco thrust, a segment of the Shiquanhe-Gerze-Ambo thrust system that bounds the north of the Gerze Basin, could also have reactivated and further elevated the Xianqian Basin (Fig. 18D). Alternatively, middle- to lower-crustal flow from the Qiangtang interior may have contributed to crustal thickening beneath the Xianqian Basin and then the Bangong-Nujiang suture basins, which likely played a key role in surface flattening (Li et al., 2024b).

However, this latter possibility would be at odds with the interpretation that lithospheric dripping drove surface subsidence, which requires strong coupling between the mantle lithosphere and the upper crust (McMillan and Schoenbohm, 2023).

6. CONCLUSIONS

The Xianqian Basin, oval-shaped with an east-west-trending long axis in the southwestern Qiangtang terrane, was filled with the 50–43 Ma Maerguo Formation. From oldest to youngest, the depositional systems of the Maerguo Formation changed from alluvial fan to fluvial, shallow lacustrine, deltaic and deep lacustrine, and finally to alluvial fan again. Most of the Maerguo Formation clastic sediments were derived from older sedimentary and volcanic rocks surrounding the basin, as suggested by bidirectional paleoflow throughout much of the basin history. A significant shift in sediment sources, from sedimentary and volcanic rocks in a wider drainage to granitoid and metamorphic rocks in a more localized drainage, occurred at ca. 45 Ma in the northern Xianqian Basin. Both the granitoid and metamorphic rocks were sourced from the north of the Xianqian Basin. The presence of abundant continental red beds and gypsum in the lower part of the Maerguo Formation indicates an

arid environment. The appearance of plant fragments, insects, bioturbation, dark-colored fine-grained clastic rocks, and other aquatic fossils in the upper part of the Maerguo Formation at ca. 43 Ma suggests less arid climatic conditions.

The Xianqian Basin developed contemporaneously with north-south crustal shortening associated with the Xianqian thrust, as evidenced by syncontractional growth strata in the northern Xianqian Basin. Since basin subsidence occurred simultaneously along both the northern and southern margins of the Xianqian Basin, as evidenced by widespread fluvial deposits of unit Em₂, we propose that lithospheric dripping was responsible for the local shortening and basin subsidence. In comparison with the Bangong-Nujiang suture basins, the paleogeographic reconstruction indicates a southward uplift pattern in the Qiangtang plateau. These lines of evidence argue for a Tibetan Plateau that was uplifted not solely in a south-to-north sequence, but also expanded southward, starting from the interior of the Qiangtang terrane.

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